

Private Delegation of (Non-)Membership Proof Updates in Cryptographic Accumulators

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Abstract

A universal, dynamic accumulator is a verifiable data structure that compresses a set of elements (e.g., unspent coins, issued public key certificates, etc.) into a succinct digest while supporting addition and deletion of elements alongside efficient proving of (non-)membership in that set. In many applications, valid (non-)membership proofs are a prerequisite to access a service (e.g., send a private payment transaction, establish a TLS connection, etc.). Typically, newly added or deleted elements necessitate updating all existing (non-)membership proofs per update. Thus, intermittently connected clients will possess invalid (non-)membership proofs whenever they reconnect. In this work, we design, implement, and evaluate algorithms for the RSA and bilinear accumulators that allow a resource-constrained client to privately delegate the updates of its (non-)membership proofs to an untrusted server. We define and prove security in a game-based framework under standard assumptions. We also study proof delegation in the batch setting. The online client algorithms are constant-time, i.e., independent of the updated set size k compared to prior $O(k)$, $O(\sqrt{k})$ works. The private delegation algorithms for membership proofs incur small concrete computational overhead for the server compared to the non-private membership proof creation algorithms, e.g., 6.99% overhead when 2^{10} elements were added in the offline phase to the RSA accumulator.

1 Introduction

Imagine a client owning an element x in a set S_t at time t , e.g., unspent coins in a cryptocurrency's ledger, public keys in a key directory, etc. Suppose that the set S_t is large, i.e., $|S_t| \gg 2^{10}$, and more importantly, that it is *dynamically changing* over time. In other words, operators can add and remove elements from the set S_t , i.e., typically $S_t \neq S_{t'}$. We assume that S_t is represented succinctly as Acc_t , i.e., as an accumulator value using a universal, dynamic cryptographic accumulator scheme [9, 51]. Suppose that the client can use a service (e.g., send a cryptocurrency transaction [55], end-to-end encryption), if it can prove with a (non-)membership proof $\pi_{t,x}(\overline{\pi_{t,x}})$ that $x \in S_t$ ($x \notin S_t$). We observe that in many applications of cryptographic accumulators (e.g., stateless blockchains [14, 24], registration-based encryption [36, 39], transparency logs [53]), clients are

running on resource-constrained devices, and are typically intermittently connected. Thus, whenever they reconnect after an offline period, they possess an outdated (non-)membership proof disabling them from accessing the desired services. Thus, the clients' goal is to possess a valid (non-)membership proof $\pi_{t,x}(\overline{\pi_{t,x}})$ corresponding to the element x at time t *without forfeiting their privacy*, i.e., releasing anything about the element x . Thus, our goal in this work is fourfold as follows.

Proof Delegation Resource-constrained clients must be able to delegate the updates of their (non-)membership proofs $\pi_{t,x}(\overline{\pi_{t,x}})$ for all time t , element x to an untrusted server, we call *proof-serving node* (PSN).

Privacy The proof-serving node should not learn anything about the element x , it updates the (non-)membership proofs for. More precisely, two (non-)membership updates should be computationally indistinguishable.

Client efficiency Ideally, the clients' computational complexity in updating their (non-)membership proofs is in $O(\text{poly}(\log(k)))$, whenever the server inserts/deletes k elements into/from the set S_t , cf. Table 1.

Public Verifiability Serving thousands of users and updating their (non-)membership proofs is a resource-intensive task. Thus, PSNs must be financially incentivized properly to serve their clients. To that end, we devise efficient, publicly verifiable proofs for (non-)membership proof updates allowing a PSN to prove that they successfully completed their computational tasks.

We only trust PSNs for availability (i.e., the server is online and does its intended job), but not for privacy. Note that there are already established efficient markets (e.g., the Golem network [40]) for verifiably outsourcing computational tasks [41].

1.1 The Necessity of Frequent Proof Updates

Recently, a growing body of literature has been developed that exposes the limits of accumulators and related cryptographic primitives, such as vector commitments. Christ and Bonneau [27] showed that there is no interesting asymptotic trade-off between the accumulator's digest size and the number of proofs that must be updated. In other words, either for n elements the accumulator digest is $O(1)$ -sized and every

membership proof must be updated or the accumulator digest is $O(n)$ -sized (*i.e.*, the entire database is stored), and thus no proofs need to be updated. The authors demonstrate that this impossibility result is universal and applies to all major authenticated data structures used in blockchains, including cryptographic accumulators, vector commitments, and authenticated dictionaries. As a concrete example, adding a single leaf to a standard Merkle tree [54] changes the root and potentially every node on the path, invalidating the witnesses/proofs for all existing leaves. This results in $O(n^2)$ total witness updates to add n items. In a follow-up work, Bonneau *et al.* [16] show that when inserting n new elements into a positive accumulator, then each membership proof needs to be updated $\omega(1)$ times. In fact, more is true, *i.e.*, any accumulator with succinct state (polylogarithmic size) must incur at least $\omega(n)$ total witness updates when adding n items. Abusalah *et al.* [1] investigate the same question of update frequency in the setting where a *single element* is added to a positive accumulator. They find that in this case, in an accumulator with n elements providing λ -bit security, essentially all proofs need to be updated, *i.e.*, $n - \text{negl}(\lambda)$ proofs are invalidated when a new element is inserted. These recent results underscore the inevitable need for PSNs to make accumulators and vector commitments practical.

1.2 Applications

We detail three applications for delegating (non-)membership proof updates privately in cryptographic accumulators.

1.2.1 Stateless Blockchains. In traditional blockchains such as Bitcoin and Ethereum, users can stay offline for a long time and simply broadcast a transaction when ready. In stateless blockchain designs [14, 24], users must constantly monitor the network to update their proofs as the blockchain state evolves. If they did not update their proofs would become stale, and they could not spend their coins. Roughly speaking, the state of a cryptocurrency system at time t is essentially the set $S_t = \{x_i\}_{i=1}^n$ consisting of all the unspent coins in the system. Each new transaction removes elements from S_t and inserts new unspent coins into S_t , updating it to a new $S_{t'}$. However, storing the entire set S_t is burdensome for smaller clients.¹ In a stateless blockchain, the full nodes store the entire set S_t , while clients only store an accumulator digest Acc_t to the S_t . Whenever clients want to spend a coin x , they can do so by issuing a valid membership proof $\pi_{t,x}$ along with other authorization material, *i.e.*, digital signatures, and other proofs. Interestingly, in a privacy-preserving cryptocurrency system, such as Zcash [7], clients issue a zero-knowledge proof-of-knowledge of owning $\pi_{t,x}$, since clients would lose their privacy if they disclose which unspent coins they own. This design motivates the need for *privately* delegating the updates of (non-)membership proofs in the stateless blockchain design context.

1.2.2 Anonymous Credentials with Revocations. In anonymous credential systems, cryptographic accumulators are commonly employed to encode the revocation state of credentials in a compact and publicly verifiable manner, enabling users to

prove membership or non-membership of a hidden revocation handle. A central practical challenge, however, is that accumulator witnesses must be updated whenever the underlying set changes, which is problematic for intermittently connected users such as mobile wallet holders. This paradigm underlies classical credential frameworks based on Camenisch–Lysyanskaya signatures [20, 21] as well as pairing-based constructions [13]. Dynamic RSA and bilinear accumulators provide succinct witnesses and scalable revocation [8, 56], but require users to update witnesses whenever the accumulator changes. Our framework extends this model by allowing users to outsource witness updates to an untrusted server while preserving unlinkability and public verifiability, thereby improving practicality for intermittently connected users in large-scale anonymous credential deployments.

1.2.3 Registration-based Encryption. Registration-Based Encryption [36, 37] (RBE) is a cryptographic primitive designed to eliminate the trusted key authority present in Identity-Based Encryption (IBE) [15] by replacing it with an untrusted “key curator” that maintains a succinct, compressed digest of all registered user public keys. This system suffers from the exact same fundamental efficiency bottlenecks as stateless blockchains, as both rely on a dynamic global commitment that changes with every newly registered user. Just as spending a coin in a stateless blockchain invalidates the inclusion proofs for other users, registering a new user in an RBE system updates the global digest, forcing all existing users to download data and update their own decryption keys to remain compatible with the new state. This is information-theoretically impossible to avoid [16], *i.e.*, users cannot simply generate a key once and go offline; instead, they face a witness update burden that scales with the number of new registrations (specifically $\Theta(n \log n / \log \log n)$), mirroring the heavy bandwidth requirements that plague stateless cryptocurrency designs.

1.3 Strawman Solutions

A straightforward idea is to simply send the (non-)membership proof $\pi_{t,x}$ ($\overline{\pi_{t,x}}$) to the PSN to update these proofs as the set S_t dynamically evolves over time. This strawman solution certainly outsources (non-)membership updates, however, at the cost of sacrificing privacy. Note that the PSN can break privacy by brute-forcing the (non-)membership verification checks, *i.e.*, for instance, for membership proofs, the PSN checks $\forall s_i \in S_t : \text{VerMem}(\text{Acc}_t, \pi_{t,x}, s_i) \stackrel{?}{=} 1$, which holds iff $s_i = x$. Assuming $|S_t| = \text{poly}(\lambda)$ in the security parameter λ , this brute-force computation can be carried out efficiently.

To preserve the privacy of the outsourced element x , one might apply fully homomorphic encryption on the outsourced (non-)membership proofs [38]. In detail, the client could homomorphically encrypt $\pi_{t,x}$ ($\overline{\pi_{t,x}}$) and send it to the PSN, who completes the (non-)membership proof update on the homomorphic ciphertext. This approach solves the problem of privately delegating accumulator proofs, but yields a less practical solution than applying our tailor-made techniques.

¹For example, the set of unspent coins in Bitcoin takes up tens of gigabytes [31].

Scheme	Accumulator	Update msg. size	Client cost	Server cost	Mem/Non-mem	Batching	Decentralized	PQ secure
[18]	Merkle tree [54]	$O(1)$	$O(1)$	$O(k \log_2(S_r))$	✓/✗	✗	✓	✓
[4]	RSA [9]	$O(k)$	$O(k)$	$O(k)$	✓/✗	✗	✓	✗
Section 5.3	RSA [9]	$O(1) \setminus O(1)$	$O(1) \setminus O(1)$	$O(k) \setminus O(k)$	✓/✓	✓	✓	✗
[65]	Bilinear [56]	$O(k)$	$O(k)$	$O(k \log_2(k))$	✓/✓	✓	✗	✗
[44]	Bilinear [56]	$O(\sqrt{k})$	$O(\sqrt{k})$	$O(k)$	✓/✗	✗	✓	✗
Section 5.4	Bilinear [56]	$O(1) \setminus O(k) \setminus O(k)$	$O(1) \setminus O(k) \setminus O(k)$	$O(k) \setminus O(k)$	✓/✓	✓	✓	✗

Table 1: Comparing our related work on privately delegating (non-)membership proofs in accumulators. The column “Mem/Non-mem” indicates that the delegation schemes support (non-)membership proofs. The column “decentralized” denotes whether the delegation schemes assume that an accumulator manager knows the trapdoor of the accumulator (e.g., group order, crs trapdoor, etc.) or not. Offline/preprocessing costs are written in parenthesis. If a scheme supports both (non-)membership proof delegations, then these protocols’ asymptotics is separated by a backslash.

1.4 Our Contributions

In this work, we provide the following three contributions.

Private accumulator proof delegation framework To the best of our knowledge, for the first time, we define private accumulator proof delegation and its security, privacy, and soundness in a game-based security framework.

Efficient constructions We construct the first efficient proof delegation protocols for the RSA [9], and bilinear accumulators [56] that achieve privacy, public verifiability, and constant client complexity.

Open-source implementation We provide the first open-source implementation of private accumulator proof delegation and evaluate its performance. Our code is publicly available at <https://github.com/GlaszBoti/private-accumulator-proof-delegation>.

Organization. This paper is organized as follows. In Section 2, we review the related work. We provide the necessary pertinent background in Section 3. We describe our system model in Section 4. We introduce our constructions in Section 5, and optimizations in Section 6. We discuss our implementation and performance evaluation in Section 7. Finally, we conclude our paper in Section 8 with open problems and future directions.

2 Related Work

2.1 Accumulators and Privacy

The anonymous credential literature [20, 22] often relies on cryptographic accumulators to store revoked credentials [19]. These works typically do not provide privacy for the witness update operation [19, 51, 60], *i.e.*, the server knows which witness it is updating. Even if they provide privacy-preserving witness update operations, they do not offer poly-logarithmic update costs for clients [4, 44] *cf.* Table 1. To the best of our knowledge, there are two private membership proof delegation schemes [18, 65] that provide a constant-time client, a necessity in real-world applications. The delegation scheme of [65] works in the accumulator manager setting for the bilinear accumulator [65], *i.e.*, incompatible with decentralized applications. The scheme of [18] for Merkle trees applies heavy recursive SNARK machinery, we wish to concretely outperform. Baldimtsi *et al.* create new accumulators that hide the

elements and set size, *i.e.*, a so-called oblivious accumulator [5]. In the applications we have in mind, the accumulated set is typically public. Barthoulot *et al.* explore delegating accumulator proofs privately to a server in a generic setting [6]. They characterize what properties a generic solution must satisfy, and they conclude that likely such a construction does not exist. In contrast, we show non-generic techniques that are tailored to the accumulator schemes at hand in a white-box manner. Thus, we can circumvent their impossibility result. Bowe and Miers devised updatable membership proofs for Merkle trees using SNARK recursion in the context of privacy-preserving cryptocurrencies [18]. However, they provided neither security proofs for their constructions nor implementation and performance evaluation. In contrast, in this work, we do not resort to heavy techniques such as proof recursion, thus, we hope to achieve better concrete performance.

2.2 Delegation of Cryptographic Computations

The study of securely delegating cryptographic computations to a potentially malicious server was initiated by Hohenberger and Lysyanskaya [43]. In particular, to our discussion, outsourcing (modular) exponentiations is the most related research direction [25, 59, 67]. These works mostly study the problem of outsourcing an exponentiation $g^u \in \mathbb{G}$, where both the base g and the exponent u must be kept secret from the helping server. Additionally, the server wants to ensure the verifiability of the output g^u . At a first glance, one may think that updating (non-)membership proofs in certain accumulators (e.g., the RSA accumulator) is “just” outsourcing an exponentiation to an untrusted server. Observe that in our setting, these techniques are not applicable, since the client does not even know the exponent u (derived from the newly added/deleted items) it wishes to raise its outdated membership proof g . An added challenge we must solve is to show that the output of the outsourced cryptographic computation is correct w.r.t. the latest accumulator digest. The typical output verifiability notion considered in the aforementioned papers is insufficient in our applications, as the server must show validity to a public value, *i.e.*, the accumulator’s latest digest.

Notation	Description
x, s_j	Element in the accumulated set
S_t	The accumulated set S at time t
n	The size of S_t , i.e., $ S_t := n$
k	The number of added elements from t to t'
l	The number of deleted elements from t to t'
Acc_t	The accumulator digest of the set S at time t
$\pi_{t,s_j} (\overline{\pi_{t,s_j}})$	(Non-)membership proof for s_j at time t
$\text{Primes}(\lambda)$	The first $2^{2\lambda}$ prime numbers in $\mathbb{N}$
$H_{\text{prime}}(\cdot)$	A hash function from $\{0, 1\}^*$ to $\text{Primes}(\lambda)$

Table 2: Notations used throughout the paper.

2.3 Private Information Retrieval

The problem of privately delegating (non-)membership proofs for cryptographic accumulators can be cast [44] as a Private Information Retrieval (PIR) problem [26]. Given a database of valid membership proofs $D_t = \{\pi_{t,s_j}\}_{s_j \in S}$, the client wants to obtain $D_t[i] := \pi_{t,s_i}$ for some index $i \in \mathcal{I}$ without disclosing the index i to the server [44]. An added challenge in our setting is that the database is dynamically changing; most PIR protocols do not support this efficiently [52]. There are a multitude of lower bounds for single-server PIR protocols. For example, a typical lower bound for PIR protocols in the information-theoretic setting states that $\Omega(n)$ bits must be communicated by the server, where n is the size of the database [42]. In the preprocessing setting, there is the inherent tradeoff of online and offline communication complexity. In particular, Corrigan-Gibbs and Kogan showed the lower bound for any preprocessing PIR, i.e., $C \cdot T \geq \tilde{\Omega}(n)$, where C denotes the bits of communication in the offline phase, and T denotes the number of bits probed of the database in the online phase [29]. In contrast, we create protocols where the client's communication and computation complexity are both $O(1)$ by exploiting the algebraic structure of the underlying database $D_t = \{\pi_{t,s_j}\}_{s_j \in S}$.

3 Background

3.1 Notations

The randomization of a group/field element x is denoted as x' . We use a multiplicative notation for the group operation in both the elliptic curve and the group of unknown order contexts. Deterministic assignment is denoted as $=$, while probabilistic assignment or uniform random sampling from a set S is denoted as $x \leftarrow S$. Our notation is summarized in Table 2.

3.2 Universal, Dynamic Accumulator Syntax

A universal, dynamic accumulator Π is a tuple of six probabilistic polynomial time (PPT) algorithms [5, 9] as follows.

$\text{Setup}(1^\lambda) \rightarrow \text{pp}$. Given a security parameter λ , this algorithm returns some public parameters pp . The public parameters pp are implicitly available to all other algorithms.

$\text{Add}(\text{Acc}_t, x) \rightarrow (\text{Acc}_{t+1}, u)$. Given an accumulator Acc_t and element x , the addition algorithm returns the new accumulator Acc_{t+1} with x added to the set of accumulated values and an update information u .

$\text{Del}(\text{Acc}_t, x) \rightarrow (\text{Acc}_{t+1}, u)$. Given an accumulator Acc_t and element x , the deletion algorithm returns the new accumulator Acc_{t+1} with x being deleted from the set of accumulated values S_t and an update information u .

$(\text{Non})\text{MemProofCreate}(\text{Acc}_t, x) \rightarrow \pi_{t,x}$. Given an accumulator Acc_t and element x , the (non-)membership creation algorithm returns a (non-)membership proof $\pi_{t,x}$.

$(\text{Non})\text{MemProofUpdate}(\text{Acc}_t, \pi_{t,x}, S_{t'}) \rightarrow \pi_{t',x}$. Given an old accumulator value Acc_t , an outdated membership proof $\pi_{t,x}$, and the latest set $S_{t'}$, the proof update algorithm returns an updated (non-)membership proof $\pi_{t',x}$.

$(\text{Non})\text{MemVer}(\text{Acc}_t, x, \pi_{t,x}) \rightarrow \{0, 1\}$. For a digest Acc_t , element x , and a (non-)membership proof $\pi_{t,x}(\overline{\pi_{t,x}})$, the (non-)membership verification algorithm outputs 1 or 0.

Definition 1 (Correctness). An accumulator scheme Π is said to be correct if all honestly generated proofs are accepted with overwhelming probability. More formally, $\forall x \in S_t$:

$$\Pr(\text{MemVer}(\text{Acc}_t, x, \text{MemProofCreate}(\text{Acc}_t, x)) = 1) \geq 1 - \text{negl}(\lambda) .$$

Definition 2 (Soundness). An accumulator scheme Π is said to be sound if all adversarially generated proofs are accepted with negligible probability. More formally, $\forall x \notin S_t$:

$$\Pr(\text{MemVer}(\text{Acc}_t, x, \text{MemProofCreate}(\text{Acc}_t, x)) = 1) \leq \text{negl}(\lambda) .$$

3.3 The RSA and Bilinear Accumulators

We recall the efficient algorithms of the RSA [9] and bilinear [56] universal dynamic accumulators in Figures 10 and 11.

3.4 Non-interactive Zero-knowledge Proofs

We recall the relevant syntax of non-interactive zero-knowledge (NIZK) proofs following [11], and for the details and exact security requirements, we refer to [11]. NIZK arguments consist of four PPT algorithms that are defined with respect to a relation generator algorithm $\mathcal{R}\text{-Gen}(1^\lambda)$ that, upon receiving some security parameter λ , outputs a polynomial time decidable relation $\mathcal{R} : \{0, 1\}^* \times \{0, 1\}^*$ for which in our case $(\phi, w) \in \mathcal{R}$, where ϕ is typically an algebraic statement in a ring $\mathbb{F}_N$ or in a finite field $\mathbb{F}_p$ and w is a valid witness for the instance.

- $\text{NIZK.Setup}(\mathcal{R}) \rightarrow (\text{crs}, \tau)$. For the relation $\mathcal{R}$, the setup produces a common reference string crs and a simulation trapdoor τ . This possibly randomized algorithm may use public coins (transparent setup) or private coins (trusted setup).
- $\text{NIZK.Prove}(\mathcal{R}, \text{crs}, \phi, w) \rightarrow \pi$. Given $(\phi, w) \in \mathcal{R}$ and the crs , the prover returns an argument π .
- $\text{NIZK.Verify}(\mathcal{R}, \text{crs}, \phi, \pi) \rightarrow \{0, 1\}$. Upon the common reference string crs , the statement ϕ , and an argument π , the verification algorithm returns 0 or 1.
- $\text{NIZK.Sim}(\mathcal{R}, \tau, \phi) \rightarrow \pi$. Using the simulation trapdoor, τ , and statement ϕ , the simulator returns π .

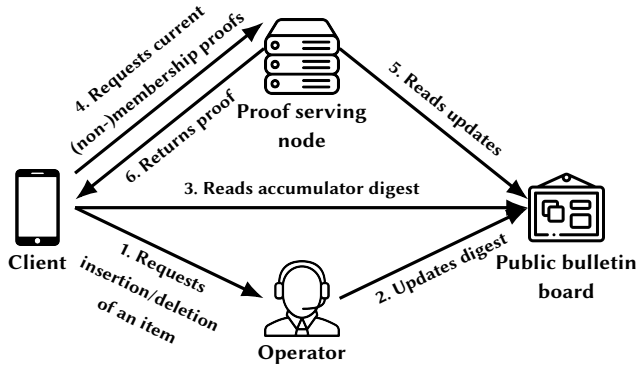


Figure 1: Our System Model. A client ① requests the insertion/deletion of an item into/from the accumulated set. Subsequently, ② the operator regularly updates the accumulator digest and publishes it to a public bulletin board (e.g., a blockchain) to reflect its current state. Naturally, ③ the client (and all other parties as well) can read the latest accumulator digest and elements from the bulletin board. After an offline period, ④ the client may request its latest valid (non-)membership proofs from a proof serving node (PSN). Finally, ⑤–⑥ the PSN serves the client with its latest (blinded) valid (non-)membership proofs.

This work builds on NIZKs for the following efficiently decidable languages in both groups of unknown order and elliptic-curve groups. All of the corresponding proofs have constant size and constant-time verification.

Discrete Logarithm equality (DLEq) [14, 23].

$$\mathcal{R}_{\text{DLEq}} = \{((g, u, h, v) \in \mathbb{G}^4); w \in \mathbb{Z} : u = g^w \wedge v = h^w\}. \quad (3.1)$$

Polynomial Exponentiation Equality (PoEEq) A variant of the DLEq proof for “exponentiating” with a polynomial, cf. [63].

$$\mathcal{R}_{\text{PoEEq}} = \{((g, u, h, v) \in \mathbb{G}^4, \phi(X) \in \mathbb{F}_p[X]; \tau \in \mathbb{F}_p) : u = g^{\phi(\tau)} \wedge v = h^{\phi(\tau)}\}. \quad (3.2)$$

4 System and Threat Models

In this work, we apply a discrete time model. A timer $t \in \mathbb{N}$ is only incremented, when an item is inserted or deleted from the accumulator with digest Acc_t , i.e., we update our timer from t to $t + 1$, and the digest to Acc_{t+1} . Next, we describe our system model that we illustrate in Figure 1.

Client We assume resource-constrained, intermittently connected clients, e.g., a browser, a cryptocurrency wallet, smart watch, etc., that want to privately delegate the updates of (non-)membership proofs for an element $x \in S$ ($x \notin S$). The client possesses a valid (non-)membership proof for an item x w.r.t Acc_t , and wishes to obtain a valid (non-)membership proof for $\text{Acc}_{t'}$, where $t' > t$. Naturally, the client does not know which elements were added or removed from the set S between time t and t' .

Public Bulletin Board There is a trusted public bulletin board (e.g., a blockchain or an independent cloud server) readable by everyone, acting as a source of truth, i.e., it stores the current accumulator digest Acc_t at all times t , enabling clients to verify (non-)membership proofs.

Operator The public bulletin board is operated by a set of incentivized parties. For instance, they are validators in the blockchain context or certificate authorities in the key transparency application. The operator appends new elements, removes necessary elements from the accumulator, updates the accumulator digest on the public bulletin board i.e., manages the accumulator.

Proof Serving Nodes Untrusted third parties, the so called proof serving nodes (PSNs), privately update the clients’ (non-)membership accumulator proofs. We envision that in practical accumulator deployments, PSNs and the operators might be the same entities, but for conceptual reasons, we separate them in our discussion.

4.1 Supporting Various Set Operations

Clients can update the accumulated set S_t in various ways depending on the application. This work focuses mainly on supporting the following two set operations.

Addition Users or the accumulator manager may add new elements to the accumulator i.e., $S_{t+1} := S_t \cup s_{n+1}$. In our target applications, this operation models the creation of new coins in stateless blockchains, or the issuance of new public key certificates in key transparency logs.

Deletion Users or the accumulator manager may remove elements from the set, i.e., $S_{t+1} := S_t \setminus s_j$ for some $s_j \in S_t$. This operation models the spending of a cryptocurrency coin, or the revocation of a public key certificate.

We admit that there are more interesting set operations that we may consider, e.g., element modification or element modification by some allowed function $f(\cdot)$, i.e., $s_{t+1,j} := f(s_{t,j})$. We leave it to future work to devise privately delegatable accumulator schemes supporting other set operations as well.

4.2 Threat Model

As we show later, our protocols offer strong privacy guarantees, i.e., the server cannot distinguish two (non-)membership proof updates for the same time interval (cf. Definition 5). However, we acknowledge that the PSN can use various *side-channel information* to breach user privacy. For example, the PSN server may take advantage of update request timing correlations, repeated queries, or other correlation attacks. We consider these leakages to be orthogonal to this work, as they can be circumvented by the appropriate use of anonymous communication tools, e.g., Tor, virtual private networks, etc.

5 Privately Delegating Accumulator Proofs

We aim to endow any universal, dynamic accumulator scheme with four new efficient algorithms.²

²Four algorithms for membership proofs, and four for non-membership proofs.

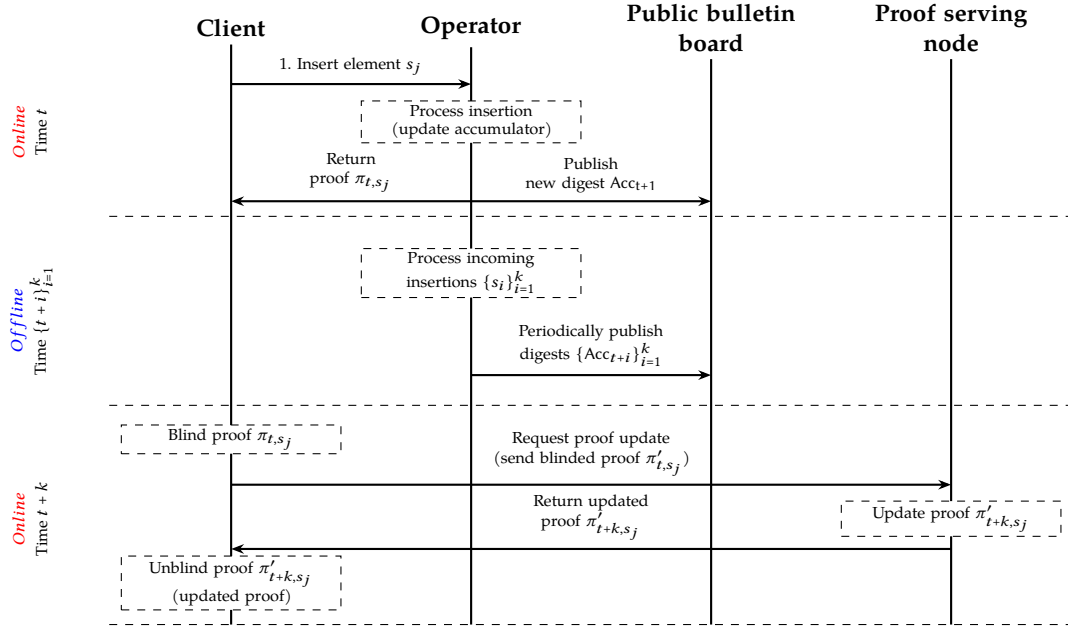


Figure 2: A sequence diagram illustrating the interactions of participants (cf. Section 4) in a privately delegatable cryptographic accumulator scheme. In a private delegation protocol for (non-)membership proof updates, at time $t + k$, the client requests a proof serving node the update of an outdated proof π_{t,s_j} for a blinded element s_j to the latest valid proof π_{t+k,s_j} .

5.1 Syntax of Privately Delegatable Cryptographic Accumulators

We introduce privately delegatable cryptographic accumulator schemes, *i.e.*, we endow universal, dynamic accumulator schemes $\mathcal{ACC} = (\text{Setup}, \text{Add}, \text{Del}, \text{MemProofCreate}, \text{MemVer}, \text{NonMemProofCreate}, \text{NonMemVer})$ with four new algorithms *i.e.*, $\Pi_{\mathcal{ACC}} = (\text{BlindProof}, \text{BlindProofUpd}, \text{VerBlindProofUpd}, \text{UnBlindProof})$ enabling to privately delegate membership proof updates in $\mathcal{ACC}$. For the sake of notational simplicity, we define our syntax and security definitions only for privately delegating *membership proofs*. It is straightforward to define them analogously for non-membership proofs as well.

Definition 3 (Privately Delegatable Accumulator). We say an accumulator scheme $\mathcal{ACC}$ has privately delegatable membership proof updates if, in addition, it also implements the following four PPT algorithms $\Pi_{\mathcal{ACC}} = (\text{BlindProof}, \text{BlindProofUpd}, \text{VerBlindProofUpd}, \text{UnBlindProof})$ where the public parameters pp of $\mathcal{ACC}$ are assumed to be implicitly given.

$\text{BlindProof}(\text{Acc}_t, x, \pi_{t,x}) \rightarrow (\pi'_{t,x}, \text{st})$. Given a membership proof $\pi_{t,x}$ for the element x at time t , this algorithm returns a blinded membership proof $\pi'_{t,x}$ and a state st .

$\text{BlindProofUpd}(\text{Acc}_t, \{x_i\}_{i=1}^k, \{y_i\}_{i=1}^l, \text{Acc}_{t'}, \pi'_{t,x}) \rightarrow (\pi'_{t',x}, \text{aux})$. For a blinded membership proof $\pi'_{t,x}$, element x and the old accumulator value Acc_t , the server can update the blinded membership proof $\pi'_{t,x}$ to a new blinded membership proof $\pi'_{t',x}$ so that it will be valid for the updated accumulator value $\text{Acc}_{t'}$ which corresponds to the digest of the set $S_{t'} := (S_t \cup \{x_i\}_{i=1}^k) \setminus \{y_i\}_{i=1}^l$.

$\text{VerBlindProofUpd}(\text{Acc}_t, \text{Acc}_{t'}, \pi'_{t,x}, (\pi'_{t',x}, \text{aux})) \rightarrow \{0, 1\}$. For the digests $\text{Acc}_t, \text{Acc}_{t'}$ it is publicly verifiable that the proof update from $\pi'_{t,x}$ to $(\pi'_{t',x}, \text{aux})$ was computed correctly. This deterministic algorithm returns a bit indicating whether the proof delegation was done correctly.

$\text{UnBlindProof}(\text{Acc}_{t'}, x, \pi'_{t',x}, \text{st}) \rightarrow \pi_{t',x}$. Given the latest accumulator digest $\text{Acc}_{t'}$, a blinded membership proof $\pi'_{t',x}$ for the element x and a state st , this algorithm returns an unblinded membership proof $\pi_{t',x}$ for $x \in S_{t'}$.

Remark 1. Note that it is necessary to blind the membership proof $\pi_{t,x}$ to $\pi'_{t,x}$, otherwise the PSN could easily link the membership proof to an item x by repeatedly calling the membership verification algorithm of the accumulator scheme. Our model is comprehensive as we allow additions and deletions in the offline period to the accumulator, *cf.* the $\text{BlindProofUpd}(\cdot)$ function. Observe that the $\text{VerBlindProofUpd}(\cdot)$ algorithm does not have private inputs, making the witness update publicly verifiable. We will omit $\mathcal{ACC}$ from the subscript of Π as it will be obvious from the context which accumulator scheme is endowed with the private delegation algorithms.

5.2 Security and Privacy requirements

We define the security and privacy requirements for privately delegatable accumulator schemes in a game-based framework. Intuitively, the delegated proof updates when honestly computed must be accepted with overwhelming probability.

Definition 4 (Correctness). Honestly run proof updates should yield valid membership proofs with overwhelming probability.

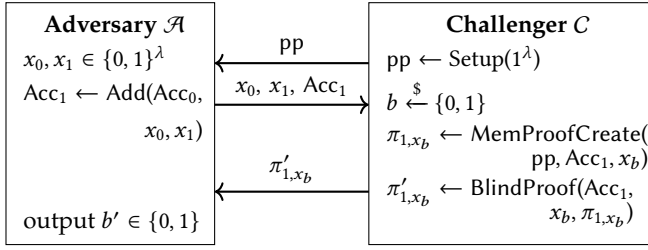


Figure 3: The blinded membership proof indistinguishability game, cf. Definition 5. $\mathcal{A}$ wins the game iff. $b = b'$.

More formally, $\forall t, t' \in \mathbb{N}, \forall \text{Acc}_t, \text{Acc}_{t'}, \forall x \in S$, we have

$$\Pr[\text{VerMem}(\text{Acc}_{t'}, x, \text{UnBlindProof}(\text{Acc}_t, \text{Acc}_{t'}, \pi_{t,x}, (\pi'_{t',x}, \text{aux}))) = 1] \geq 1 - \text{negl}(\lambda),$$

where

$$\pi'_{t,x} \leftarrow \text{BlindProof}(\text{Acc}_t, x, \pi_{t,x}), \\ (\pi'_{t',x}, \text{aux}) \leftarrow \text{BlindProofUpd}(\text{Acc}_t, \{x_i\}_{i=1}^k, \{y_i\}_{i=1}^l, \text{Acc}_{t'}, \pi'_{t,x}).$$

Intuitively, the blinded (non-)membership proofs should not leak anything about neither the element x nor the unblinded (non-)membership proof.

Definition 5 (Blinded Membership Proof Indistinguishability). A privately delegatable accumulator scheme has blinded membership proof indistinguishability if $\forall \lambda, \forall$ PPT adversary $\mathcal{A}$, $\exists \text{negl}(\cdot)$ s.t. the advantage of the adversary in the indistinguishability game (cf. Figure 3) is $\text{negl}(\lambda)$. More formally,

$$\Pr[\mathcal{G}_{\Pi}(\mathcal{A}) = 1] \leq \frac{1}{2} + \text{negl}(\lambda). \quad (5.1)$$

We also introduce a relaxed notion of blinded membership proof indistinguishability. In this relaxed notion, the adversary cannot link the blinded membership proof among *exponentially many* accumulated elements with non-negligible probability.

Definition 6 (Weak Blinded Membership Proof Indistinguishability). Let $x \in_R \{0, 1\}^\lambda$ and $\pi_{t,x}$ a valid membership proof x at time t . Moreover, we let $\pi'_{t,x} := \text{BlindProof}(\text{pp}, \pi_{t,x})$. Then, we say that a privately delegatable accumulator scheme Π satisfies weak blinded membership proof indistinguishability if $\forall (\lambda, t) \in \mathbb{N}^2, x \in_R \{0, 1\}^\lambda$, and $\forall \mathcal{A}$ PPT, there exists a $\text{negl}(\cdot)$ function such that the following holds:

$$\Pr[\mathcal{A}(\text{pp}, \pi'_{t,x}) = x] \leq \text{negl}(\lambda). \quad (5.2)$$

In Section 5.3.2, we will see a private delegation protocol that satisfies weak blinded membership proof indistinguishability, but it does not satisfy the stronger notion of blinded membership proof indistinguishability.

Next, we introduce the notion of delegation soundness. Intuitively, a malicious PSN should not be able to produce a publicly verifiable convincing proof update aux such that later the unblinded (non-)membership proof is *not* verified by the accumulators (non-)membership verification algorithms.

BlindMemProof($\text{pp}, \pi_{t,x}$):

1. $\text{st} := r \xleftarrow{\$} \mathbb{Z}_{2\lambda}$
2. $\pi'_{t,x} := \pi_{t,x} \cdot g^r$
3. **return** $(\pi'_{t,x}, \text{st})$

BlindMemProofUpd($\text{pp}, S_t, \text{Acc}_t, \{x_i\}_{i=1}^k, \{y_i\}_{i=1}^l, \text{Acc}_{t'}, \pi'_{t,x}$):

1. $S_{t'} = (S_t \cup \{x_i\}_{i=1}^k) \setminus \{y_i\}_{i=1}^l$
2. $P_T \leftarrow \prod_{i=1}^k x_i, P_R \leftarrow \prod_{i=1}^l y_i$
- 3a. $\Delta \leftarrow P_T \cdot (P_R^{-1} \bmod |\mathbb{G}|)$
- 3b. $\Delta \leftarrow P_T (\in \mathbb{Z})$ // assuming no deletion
4. $A = (\pi'_{t,x})^\Delta, B = g^\Delta$
5. $\pi_1 \leftarrow \text{NIZK.Prove}(\mathcal{R}_{\text{DLEq}}, (\pi'_{t,x}, A, \text{Acc}_t, \text{Acc}_{t'}), \Delta)$
6. $\pi_2 \leftarrow \text{NIZK.Prove}(\mathcal{R}_{\text{DLEq}}, (g, B, \pi'_{t,x}, A), \Delta)$
7. $\pi'_{t',x} \leftarrow \{A, B\}, \text{aux} \leftarrow \{\pi_1, \pi_2\}$
8. **return** $(\pi'_{t',x}, \text{aux})$

VerBlindMemProofUpd($\text{pp}, \text{Acc}_t, \text{Acc}_{t'}, \pi'_{t,x}, \pi'_{t',x}, \text{aux}$):

1. $\{\pi_1, \pi_2\} \leftarrow \text{aux}, \{A, B\} \leftarrow \pi'_{t',x}$
2. $b_1 := \text{NIZK.Verify}(\mathcal{R}_{\text{DLEq}}, (\pi'_{t,x}, A, \text{Acc}_t, \text{Acc}_{t'}), \pi_1)$
3. $b_2 := \text{NIZK.Verify}(\mathcal{R}_{\text{DLEq}}, (g, B, \pi'_{t,x}, A), \pi_2)$
4. **return** $b_1 \wedge b_2$

UnMemBlindProof($\text{Acc}_{t'}, \text{st}, \pi'_{t',x}$):

1. $\{A, B\} \leftarrow \pi'_{t',x}, r \leftarrow \text{st}$
2. $\pi_{t',x} := A/B^r$
3. **return** $\pi_{t',x}$

Figure 4: The private delegation algorithms for membership proof updates in the RSA accumulator. Step 3a of **BlindProofUpd**($\cdot$) (typeset in blue) can be vastly sped up if the PSN knows the trapdoor of the group of unknown order, i.e., its factorization in case of an RSA instantiation. This is reasonable when the PSN is also the accumulator manager.

Definition 7 ((Computational) Delegation Soundness). A privately delegatable accumulator scheme Π_{ACC} is computationally delegation sound if $\forall \lambda, \forall$ PPT adversary $\mathcal{A}$, $\forall x \in S_t, (t, t') \in \mathbb{N}$, there exists a $\text{negl}(\cdot)$ function s.t.,

$$\Pr \left[\text{MemVer}(\text{UnBlindProof}(\text{Acc}_{t'}, \text{st}, \pi'_{t',x})) = 0 \mid \text{VerBlindProofUpd}(\text{pp}, \text{Acc}_t, \text{Acc}_{t'}, \pi'_{t,x}, \pi'_{t',x}, \text{aux}) = 1 \right] \leq \text{negl}(\lambda).$$

Definition 8 (Private Delegation of Membership Proof Updates). A universal, dynamic accumulator scheme $\mathcal{ACC}$ has a secure, private delegation scheme for membership proof updates if $\Pi_{\text{ACC}} = (\text{BlindProof}, \text{BlindProofUpd}, \text{VerBlindProofUpd}, \text{UnBlindProof})$ satisfies correctness, delegation soundness, and (weak) blinded membership proof indistinguishability.


```

BlindNonMemProof(pp, x):
  1. if  $x \in S_t$ : return  $\perp$ 
  2.  $st := q \xleftarrow{\$} \text{Primes}(1^\lambda)$ 
  3.  $\pi_{t,x}' := x \cdot q$ 
  4. return  $(\pi_{t,x}', st)$ 

BlindNonMemProofUpd(pp,  $\text{Acc}_t, \{x_i\}_{i=1}^k, \{y_i\}_{i=1}^l, \text{Acc}_{t'}, \overline{\pi_{t,x}'})$ :
  1.  $s^* := \prod_{s \in S_{t'}} s \bmod |\mathbb{G}|$ 
  2.  $\{a, b \bmod |\mathbb{G}|\} \leftarrow \text{Bezout}(s^*, \overline{\pi_{t,x}'})$ 
  3.  $\overline{\pi_{t',x}'} \leftarrow \{a, B := g^b\}, \text{aux} \leftarrow \emptyset$ 
  4. return  $(\overline{\pi_{t',x}'}, \text{aux})$ 

VerBlindNonMemProofUpd(pp,  $\text{Acc}_{t'}, \overline{\pi_{t,x}'}, \overline{\pi_{t',x}'}, \text{aux}$ ):
  1.  $\{a, B\} \leftarrow \overline{\pi_{t',x}'}$ 
  2.  $y \leftarrow \overline{\pi_{t,x}'}$ 
  3. return 1 if  $g = \text{Acc}_{t'}^a B^y$ 

UnBlindNonMemProof( $\text{Acc}_{t'}, st, \overline{\pi_{t',x}'}$ ):
  1.  $\{a, B\} \leftarrow \overline{\pi_{t',x}'}$ 
  2.  $q \leftarrow st$ 
  3.  $B' := B^q$ 
  4. return  $\overline{\pi_{t',x}'} \leftarrow \{a, B'\}$ 

```

Figure 5: The private delegation algorithms for updating non-membership proofs in the RSA accumulator. The steps 1 and 2 of $\text{BlindNonMemProofUpd}(\cdot)$ (typeset in blue) can be largely sped up if the PSN knows the trapdoor of the group of unknown order, i.e., its factorization in case of an RSA instantiation. This is reasonable when the PSN is also the accumulator manager.

5.3 Privately Delegating (Non-)Membership Proofs in the RSA accumulator

5.3.1 Membership proofs. Due to the RSA accumulator's multiplicative structure, a simple $\pi'_{t,x} = \pi_{t,x} \cdot g^r$ randomization suffices for the blinded membership proof, while enabling efficient blinded proof updates. The server computes $\Delta = \prod_{i=1}^k x_i \in \mathbb{Z}$ and the exponentiation $\pi'_{t,x} := (\pi_{t,x})^\Delta$ to obtain the updated blind membership proof. Finally, the server shows consistency between $(\text{Acc}_t, \text{Acc}_{t'})$ and $(\pi'_{t,x}, \pi_{t',x})$ by creating a discrete logarithm equality proof. We describe the four algorithms for our privately delegatable RSA accumulator scheme in Figure 4.

Theorem 1 (RSA accumulator is privately delegatable for membership proofs). Our construction in Figure 4 yields a secure, privately delegatable membership proof update scheme for the RSA with element additions and deletions under the strong RSA assumption and knowledge soundness of Figure 12.

The proof can be found in Appendix C.1.

5.3.2 Non-membership proofs. Now, we describe the four efficient algorithms that allow a client to privately delegate the updates of their RSA accumulator non-membership proofs, cf. Figure 5. The intuition is that due to the hardness of factoring, for a non-element x , i.e., $x \notin S$, $x \cdot q$ hides x for appropriately large primes. Although in many RSA accumulator applications, elements are hashed to smaller primes, i.e., of size 256-bits, thus, the proposed protocol requires the use of an H_{Primes} with a larger image space for privacy reason. This is because factoring a 512-bit semiprime is considered highly practical [64].

Remark 2. The implicit assumption in the protocol presented in Figure 5 is that the size of the set clients want to update their non-membership proofs for (denote this set as S' s.t. $S' \cap S = \emptyset$), has an exponential size in λ . In the stateless blockchain application of Section 1.2.1, elements are coin identifiers hashed to 256-bit primes, giving an effectively $O(\lambda)$ -size element space; the weak notion suffices. In contrast, for certificate revocation lists with a bounded registry, this construction should not be used without additional precautions. Note, if $|S'| \in \text{poly}(\lambda)$, then it is trivial to break weak blinded non-membership proof indistinguishability: the server computes $\forall x' \in S' : \gcd(x', \overline{\pi_{t,x}'})$. If $\gcd(x', \overline{\pi_{t,x}'}) \neq 1$, the server found the element x' for which the clients obtained up-to-date non-membership proofs. Thus, this non-membership proof update delegation cannot satisfy blinded non-membership proof indistinguishability (cf. Definition 5), however, as we show now, it satisfies weak blinded non-membership proof indistinguishability, cf. Definition 6.

Theorem 2 (The RSA accumulator is privately delegatable for non-membership proofs). Our construction in Figure 5 gives a secure, privately delegatable non-membership proof update scheme for the RSA accumulator with element additions and deletions under strong RSA and factoring assumptions in the random oracle model (ROM).

The proof can be found in Appendix C.2.

5.4 Privately Delegating (Non-)Membership Proofs in Bilinear Accumulators

Next, we make the bilinear accumulator privately delegatable. Note that the following bilinear delegation protocols do not support element deletions. We leave this gap to future work.

5.4.1 Membership proofs. The high-level idea of our protocol for privately delegating membership proofs in bilinear accumulators is as follows. Recall, in a bilinear accumulator $\forall s_j \in S \Leftrightarrow (X - s_j) | f(X)$, i.e., $f(X) := \prod_{i=1}^n (X - s_i)$. Thus, the membership proof for an element s_j is the KZG commitment to the quotient polynomial $q(X) := \frac{f(X)}{X - s_j}$. Our idea is that the client sends a randomized common reference string crs' to the PSN, where each monomial in crs' is additionally raised to $r q(\tau)$, i.e., $\text{crs}' := \{\text{KZG.Com}(r q(X) X^i)\}_{i=0}^k$, where τ is the trapdoor of the KZG CRS and r is a randomizer field element sampled by the client. The PSN uses crs' in the update algorithm to commit to the updated membership proof for the element s_j , i.e., $q'(X) = \frac{f'(X)}{(X - s_j)} = q(X) \prod_{i=1}^k (X - s_i)$. The crs' allows the PSN to update the membership proof supporting up to k newly

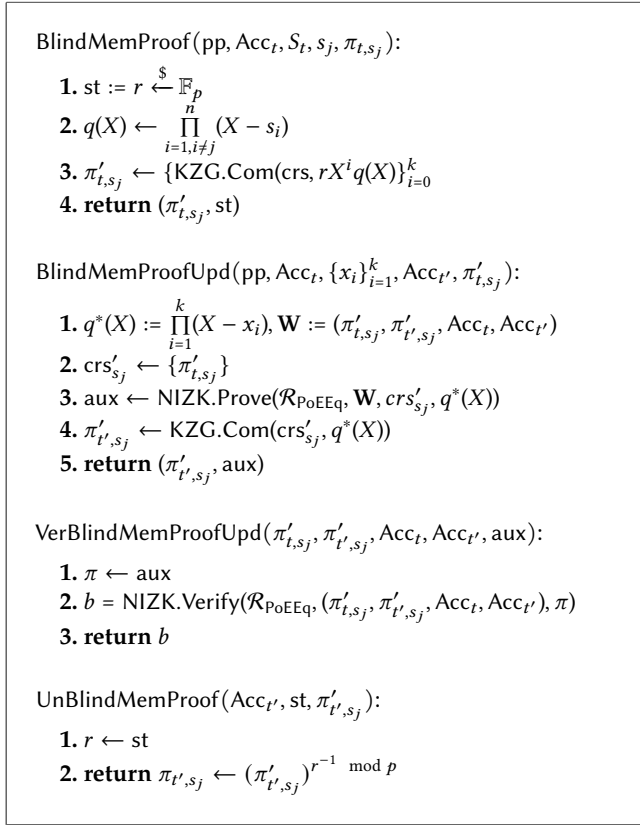


Figure 6: The delegation algorithms for privately updating membership proofs in the bilinear accumulator.

added elements. Finally, the client derandomizes the updated membership proof to obtain the latest valid membership proof for s_j. The private delegation protocol is detailed in Figure 6.

Preprocessing. At first glance, BlindMemProof(·) algorithm imposes an $O(k)$ online computational cost on the client (cf. Figure 6) to generate the blinded reference string crs_{s_j}', where k is the batch size of newly accumulated elements. However, we observe that the preparation of crs_{s_j}' is independent of the update elements {x_i}_{i=1}^k. Consequently, the client can compute this blinded CRS during an *offline, preprocessing* phase. In the online phase—when the accumulator transitions from t to t'—the heavy $O(k)$ polynomial commitment is entirely delegated to the PSN. The client's online computational cost is thereby reduced to verifying a constant-size PoEEq NIZK ($\mathcal{R}_{\text{PoEEq}}$) and performing a single unblinding exponentiation. This offline-online decoupling achieves a strictly $O(1)$ online computational complexity for the client, making the delegation protocol highly practical for resource-constrained devices.

Theorem 3 (Bilinear accumulator is privately delegatable). The construction in Figure 6 is a secure, privately delegatable membership proof update scheme for the bilinear accumulator,

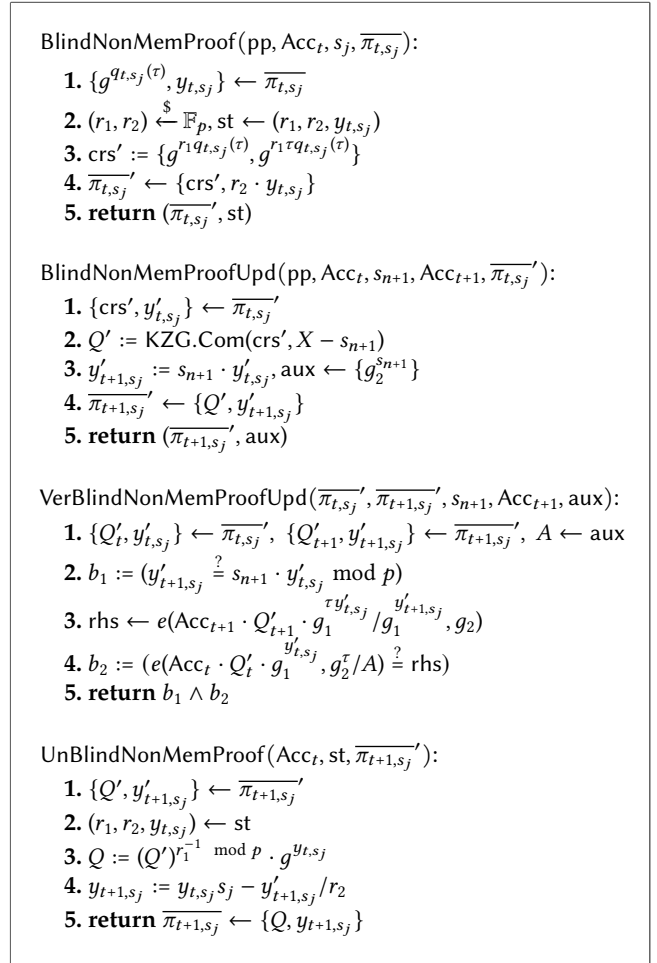


Figure 7: The private delegation algorithms for updating *non-membership* proofs in the bilinear accumulator after a single element s_{n+1} is added to the accumulated set S_t. If multiple elements are added in the offline period, then the PSN and the client repeat this protocol for each added element.

under q-Strong Diffie-Hellman (q-SDH) assumption in the algebraic group model and ROM.

The proof can be found in Appendix C.3.

5.4.2 Non-membership proofs. Unlike the membership case, the following protocol is interactive and requires k rounds between the client and the PSN – one per added element – making it suitable only for small offline update sizes. Recall that in the bilinear accumulator if s_j ∉ S_t ⇒ 0 ≠ (s_t(X) mod (X - s_j)). Thus, s_t(X) = q_{t,s_j}(X)(X - s_j) + y_{t,s_j}, where y_{t,s_j} ≠ 0. That is, for s_j at time t we have: π_{t,s_j}' := (KZG.Com(q_{t,s_j}(X), y_{t,s_j})) [30].

One may be tempted to take a similar approach as in Figure 5, i.e., using the PSN as a polynomial “Bézout-oracle”: the PSN could evaluate Bezout(s_t(X), ·) for cleverly crafted input polynomials. This approach does not appear to be fruitful, as

no matter how one blinds the desired query polynomial $(X - s_j)$ it seems difficult to obtain a meaningful protocol. For example, masking with a random r , i.e., $r(X - s_j)$ seems to be at odds not only with privacy but also correctness, as $\text{Bezout}(s_t(X), \cdot)$ is non-linear. Masking $(X - s_j)$ with higher-degree polynomials is pointless as one can efficiently factor polynomials over finite fields due to Berlekamp [10]. Instead of the previous approach, let us analyze how $(q_{t,s_j}(X), y_{t,s_j})$ must be updated after a single element s_{n+1} is added to the accumulator. That is,

$$\begin{aligned} s_{t+1}(X) &= s_t(X)(X - s_{n+1}) = (q_{t,s_j}(X)(X - s_j) + y_{t,s_j})(X - s_{n+1}) = \\ &= q_{t+1,s_j}(X)(X - s_j) + y_{t+1,s_j}. \end{aligned}$$

The polynomial division computation (i.e., $y_{t,s_j}(X - s_{n+1})/(X - s_j) = y_{t,s_j} \wedge (y_{t,s_j}(X - s_{n+1}) \bmod (X - s_j)) \equiv (y_{t,s_j}s_j - y_{t,s_j}s_{n+1}))$) yields the following recurrence relation for updating non-membership proofs in bilinear accumulators after adding a single element to the accumulated set.

$$q_{t+1,s_j}(X) = q_{t,s_j}(X)(X - s_{n+1}) + y_{t,s_j}, \quad (5.3)$$

$$y_{t+1,s_j} = y_{t,s_j}s_j - y_{t,s_j}s_{n+1}. \quad (5.4)$$

The recurrence relations above suggest the following private delegation protocol for the bilinear accumulator. The PSN server and the client can engage in an interactive protocol during the update process for each element s_{n+1} added to the set S_t . In each round of the non-membership update protocol, the client sends a very short randomized crs' allowing the PSN to KZG-commit to a randomized version of $q_{t+1,s_j}(X)$. Similarly, the server can compute and return a randomized version of y_{t+1,s_j} . At the end of the protocol, the client can verify the correctness of the final received non-membership proof elements. The full protocol is detailed in Figure 7.

Theorem 4 (Bilinear accumulator is privately delegatable for non-membership proofs). Our construction in Figure 7 yields a secure, privately delegatable non-membership proof update scheme for the bilinear accumulator with additions under the q -SDH assumption in the ROM.

The proof can be found in Appendix C.4.

6 Batching and Other Optimizations

6.1 Client-side Batching of Private Delegation

In this setting, the client owns a batch of elements $B := \{s_i\}_{i=1}^m \subseteq S_t$ and wants to privately update a batch (non-)membership proof $\pi_{t,B}(\overline{\pi_{t,B}})$ corresponding to B .

6.1.1 RSA batch membership proof updates. An RSA batch membership proof update for a batch B can be computed almost in the same fashion as in Figure 4. However, one needs to slightly modify the $\text{BlindProof}(\cdot)$ algorithm semantically, as it must now blind a batch membership proof $\pi_{t,B} := g^{\prod_{s_j \in S \setminus B} s_j}$, instead of a single membership proof, cf. Figure 14.

6.1.2 RSA batch non-membership proof updates. A similar idea also works, cf. Figure 14, for updating a batch non-membership proof. Note that in this case, the batch size is leaked to the PSN, which, in certain applications, might be untenable.

6.1.3 Bilinear batch membership proof updates. Observe that it is possible to adapt our membership proof delegation to the batch setting for the bilinear accumulator. In particular, we only need to modify the $\text{BlindProofUpd}(\cdot)$ function, Figure 14. Crucially, the PSN cannot know whether it is updating a membership proof for a single element s_j or for a larger batch B .

6.2 Server-side Batching of Private Delegation

Imagine that there are numerous querying clients (denoted Q) who request proof updates for the same time period (i.e., from t to t') for blinded membership proofs $\forall s_i \in Q : \pi'_{t,s_i}$. At a high-level, in both accumulators (cf. Figures 4 and 6), the server must compute $\forall s_i \in Q : (\pi'_{t,s_i})^\Delta$ for each queried element s_i , for some public exponent Δ , derived from the updated elements. The server's goal is to compute all updated proofs efficiently e.g., in sublinear time in $|Q|$. However, as we show next, asymptotic server-side batching is impossible in the generic group model [62]. This means that PSN operators cannot offer economies of scale in the naïve sense, which directly affects the business model of running a PSN, cf. Section 7.4.

Lemma 5 (Linear lower bound for component-wise exponentiation in the generic group model). Let $\mathbb{G}$ be a cyclic group of prime order p in the generic group model. Let $g_1, \dots, g_n \in \mathbb{G}$ be independent random group elements and let $x \in \mathbb{Z}_p$ be known. Any generic algorithm that outputs $(g_1^x, g_2^x, \dots, g_n^x)$ must perform at least n oracle queries. In particular, the generic complexity of this computational task is $\Omega(n)$.

We present the proof of this lemma in Appendix C.5.

6.3 Batched Verification Proof

After the previous negative result, now we want to speed up the verification cost of m aux proofs issued to attest the correctness of m private proof delegation tasks. In particular, consider a client who blinds m membership proofs independently but the PSN reuses the same update exponent Δ and produces a single NIZK proof that all blinded membership proofs were updated consistently with the accumulator transition.

6.3.1 RSA accumulator. Concretely, for elements $\{x_j\}_{j=1}^m \subseteq S_t$ with membership proofs $\{\pi_{t,x_j}\}_{j=1}^m$, the client samples $\{r_j\}_{j=1}^m \xleftarrow{\$} \mathbb{Z}_{2^\lambda}$ and sends blinded membership proofs

$$\pi'_{t,x_j} := \pi_{t,x_j} \cdot g^{r_j} \quad \text{for all } j \in [m],$$

to the PSN. Given the update $(S_t, \text{Acc}_t) \rightarrow (S_{t'}, \text{Acc}_{t'})$ with exponent Δ (computed as in the base scheme), the PSN computes

$$A_j := (\pi'_{t,x_j})^\Delta \quad \text{for all } j \in [m], \quad B := g^\Delta.$$

To efficiently prove correctness, the PSN and the client derive random Fiat-Shamir challenges $\alpha_j := H(\text{Acc}_t, \text{Acc}_{t'}, B, \{\pi'_{t,x_i}\}_{i=1}^m, \{A_i\}_{i=1}^m, j)$ and aggregate the instances into $X := \prod_{j=1}^m (\pi'_{t,x_j})^{\alpha_j}$ and $Y := \prod_{j=1}^m A_j^{\alpha_j}$. The PSN then returns $\{A_j\}_{j=1}^m, B$ together with a constant-size DLEq proof for the aggregated relation

$$\exists \Delta : Y = X^\Delta \wedge \text{Acc}_{t'} = \text{Acc}_t^\Delta \wedge B = g^\Delta.$$

Private Delegation Protocol	$ \pi'_{t,x} $	$ \text{aux} $	BlindProof	BlindProofUpd	VerBlindProofUpd	UnBlindProof
Merkle tree membership [18]	$O(\log_2^2(kd)) \mathbb{G} $	$\emptyset$	N/A	$\approx k(d \mathbb{Z}_p + d \mathbb{G} + H)$	$O(\log_2(kd))\mathbb{G}$	N/A
RSA membership (Figure 4)	$ \mathbb{G}_U $	$4 \mathbb{G}_U + 2 \mathbb{Z}_\lambda $	$\mathbb{G}_U$	$2k\mathbb{Z}_\lambda + 6\mathbb{G}_U + 2H_{\text{Primes}}$	$8\mathbb{G}_U + 2H_{\text{Primes}}$	$\mathbb{G}_U$
RSA non-membership (Figure 5)	$ \mathbb{Z}_\lambda $	$\emptyset$	$\mathbb{Z}_\lambda$	$k\mathbb{Z}_\lambda + \mathbb{G}_U$	$2\mathbb{G}_U$	$\mathbb{G}_U$
Bilinear membership (Figure 6)	$(k+1) \mathbb{G} $	$4 \mathbb{G} + \mathbb{Z}_p $	$(k+1)\mathbb{G}$	$(4k+2)\mathbb{G} + (2k+1)\mathbb{Z}_p + H$	$3P$	$\mathbb{G} + \mathbb{Z}_p$
Bilinear non-membership (Figure 7)	$2k \mathbb{G} + k \mathbb{Z}_p $	$k \mathbb{G}_2 $	$2k\mathbb{G} + k\mathbb{Z}_p$	$2k\mathbb{G} + k\mathbb{Z}_p + k\mathbb{G}_2$	$2kP + 3k\mathbb{G} + k\mathbb{Z}_p$	$2k\mathbb{G} + 3k\mathbb{Z}_p$

Table 3: Theoretical performance of our private (non-)membership proof update delegation algorithms for the RSA and bilinear accumulators. $\mathbb{G}, \mathbb{G}_U, P, H, \mathbb{Z}_\lambda$ denote elliptic curve, unknown order group operations (*i.e.*, RSA, class groups), pairing, hashing and finite field operations, respectively. Recall, k denotes the #elements added to S_t in the offline phase and d , the depth of the Merkle tree, *i.e.*, $d = O(\log_2(|S_t|))$. For this comparison, we instantiated [18] with the Nova folding scheme and Spartan decider SNARK [50]. Note that [18] provides no public implementation, precluding direct empirical comparison.

The client computes X and Y , verifies the DLEq proof, and then unblinds each updated blinded witness locally via

$$\pi'_{t,x_j} := A_j / B^{r_j},$$

recovering the updated RSA-accumulator membership proof $\pi'_{t,x_j} = \pi_{t,x_j}^\Delta$ for every $j \in [m]$. By aggregating the relations, the aux proof size is reduced from $O(m)$ to strictly $O(1)$ group/-field elements. The soundness for this aggregated verification relies on the *low-order assumption* in the underlying group of unknown order, ensuring that adversarial error terms cannot cancel out by wrapping around a small-order subgroup [14].

6.3.2 Bilinear accumulator. In the bilinear setting, naively verifying m independent PoEEq proofs requires the client to compute m separate pairing equations. Given that the standard PoEEq relation (*cf.* Figure 12) uses an instance-dependent Fiat-Shamir challenge α_j that appears inside the $\mathbb{G}_2$ argument of the pairing (*i.e.*, $g_2^{\tau+\alpha_j}$), these pairings cannot be directly batched. To overcome this, we modify the PoEEq protocol to use a *shared* challenge for the entire batch. First, the PSN and the client derive a single global challenge $\alpha := H(\{a_j, f_j\}_{j=1}^m) \in \mathbb{F}_p^*$. For each instance $j \in [m]$, the PSN computes the quotient polynomial $h_j(X)$ such that $f_j(X) = (X + \alpha)h_j(X) + \beta_j$, and sends the proof elements $Q_j := g_1^{h_j(\tau)}$ to the client. Because α is shared, the $\mathbb{G}_2$ pairing argument $g_2^{\tau+\alpha}$ is now identical across all m verification equations. This thus enables the client to verify the entire batch using a random linear combination. In particular, the client derives random Fiat-Shamir weights $\gamma_j := H'(\alpha, \{Q_i\}_{i=1}^m, j) \in \mathbb{F}_p^*$ and computes the aggregated elements in $\mathbb{G}_1$:

$$Q_{\text{agg}} := \prod_{j=1}^m Q_j^{\gamma_j} \quad \text{and} \quad A_{\text{agg}} := \prod_{j=1}^m (a_j \cdot g_1^{-\beta_j})^{\gamma_j}.$$

The client then accepts the batch if and only if the following single pairing equation holds: $e(Q_{\text{agg}}, g_2^{\tau+\alpha}) \stackrel{?}{=} e(A_{\text{agg}}, g_2)$.

7 Implementation and Performance Evaluation

7.1 Theoretical Performance

Observe that all client algorithms in the RSA (non-)membership delegation protocols, *i.e.*, BlindProof($\cdot$), VerBlindProofUpd($\cdot$), and UnBlindProof($\cdot$) run in $O(1)$ time (*cf.* Table 3), which is independent of the updated set size as desired in real-world applications. The linear $O(k)$ computational cost of BlindProof($\cdot$) in the

bilinear membership delegation protocol could be offloaded to a preprocessing phase. Unsurprisingly, the asymptotic cost of the PSN server is $O(k)$ in all our private delegation protocols.

7.2 Our Implementation

In the publicly available implementation³ of our protocols, all code is implemented in Rust, using the Arkworks library [3] and the Zengo class group library [28]. The criterion-rs crate was used for all benchmarks. We chose the security parameter as $\lambda = 128$. Following NIST guidelines, this amounts to using moduli N of bit-length of 3072 [57]. Following the work of Dobson *et al.* [32], we chose the discriminant Δ for class groups of imaginary quadratic fields to be 3400 bits long, providing λ bit security for class groups [33]. Accordingly, we chose the BLS12-381 curve that is currently known to provide roughly 128-bit security [17]. Interestingly, the Rust num_bigint library always chooses the most efficient integer multiplication algorithm (*i.e.*, textbook, Karatsuba [46], or Toom-Cook multiplication algorithms [49]) depending on the input integer sizes. Similarly, division is implemented using Knuth’s algorithm; when dividing by linear factors, we employ synthetic division [34]. Our hash-to-prime function $H_{\text{Prime}}(\cdot)$ is implemented with the image space of the first $2^{2\lambda}$ prime numbers to provide a collision resistance of λ -bits (and λ -bit soundness in the applied NIZKs compiled with the Fiat-Shamir heuristic [35]). Applying the prime number theorem, we set the image space of $H_{\text{Prime}}(\cdot)$ to $\{0, 1\}^{2\lambda+\log(2\lambda)}$, *i.e.*, in our case $\{0, 1\}^{262}$. We instantiate $H_{\text{Prime}}(\cdot)$ using a deterministic hash-and-increment construction, starting from $H_{\text{SHA256}}(\cdot)$ we test the primality of odd candidates with the Miller-Rabin probabilistic primality testing algorithm [58] using 64 iterations. All experiments are run on a consumer laptop with 16 GB RAM, and an Apple M2 processor with 8 cores. Each experiment was run 50 times.

7.3 Performance Evaluation

7.3.1 Runtime analysis. In Figure 8a, we measure the total runtime of private (non-)membership proof updates as the set S_t grows. Our private delegation algorithms for the bilinear accumulator significantly outperform other private membership update algorithms, achieving at least 2× faster runtime when

³See: <https://github.com/GlaszBoti/private-accumulator-proof-delegation>.

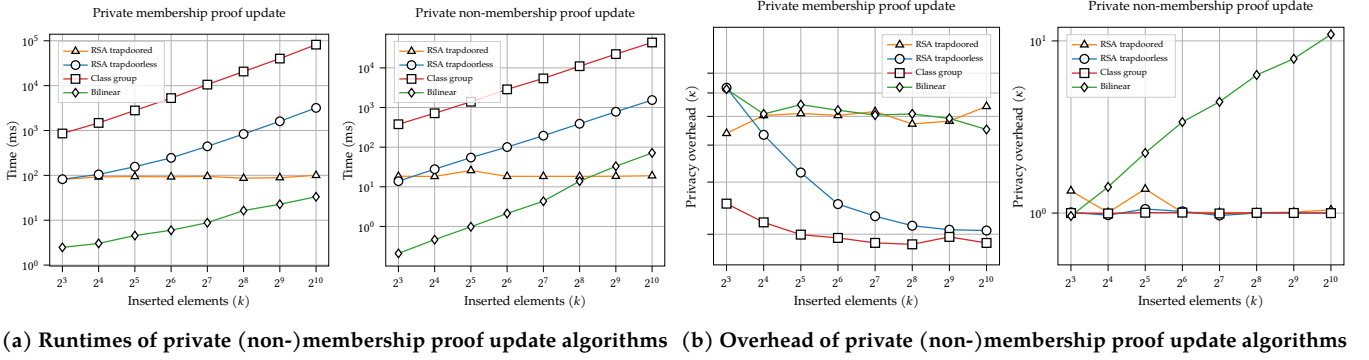


Figure 8: PSN running times in the (non-)membership proof update algorithms (Blind(Non)MemProofUpd($\cdot$)). Private update overhead is defined as the ratio of the running times of the Blind(Non)MemProofUpd($\cdot$) and (Non)MemCreate($\cdot$) algorithms.

adding up to 1024 elements. Similarly, for non-membership proof updates, the bilinear accumulator consistently outperforms every RSA accumulator instantiation. Interestingly, the trapdoor setting of RSA (*i.e.*, when the factorization is known to the server) achieves constant-time, *i.e.*, $O(1)$ updates. Figure 15 investigates which cryptographic operations (*e.g.*, hashing, exponentiation (multi-scalar multiplication), NIZK proof generation, etc.) constitute a bottleneck for the server during private proof update delegation.

7.3.2 Privacy Overhead. In Figure 8b, we evaluate the computational overhead incurred by our private (non-)membership proof update algorithms as the size of the updates grows. Formally, we define the privacy overhead as

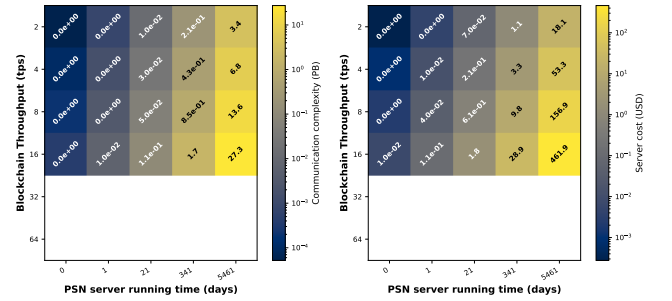
$$\kappa(k) = \frac{t(\text{Blind(Non)MemProofUpd}(\cdot, k))}{t(\text{(Non)MemProofCreate}(\cdot, k))}, \quad (7.1)$$

where $t(\cdot, k)$ denotes the running time of an algorithm when k elements were added to the accumulator in the offline phase. Intuitively, a client can request a fresh (non-)membership proof from the server without privacy or use private proof updates that may incur a performance overhead for the PSN.

We observe that the RSA accumulator’s private proof delegation algorithms’ overhead $\kappa(\cdot)$ quickly converges to 1 with the notable exception of the trapdoor RSA instantiation. In contrast, the bilinear accumulator incurs a high privacy overhead attributed to the already efficient proof creation and interactive design of non-membership proof updates. Overall, the data demonstrate that the RSA-based instantiation provides the lowest overhead for (non-)membership proof updates.

7.4 Case Study: PSN for Stateless Blockchains

We study the communication and operational costs of running a PSN server for the envisioned stateless blockchain use case, *cf.* Figure 9. At the time of writing, the hourly running cost of a t3.medium AWS instance with 8 threads is $\text{cost} = 0.052$ USD/hour [2]. In Bitcoin, the coin age distribution (the time elapsed between the creation and spending of a UTXO) follows a power-law distribution, *i.e.*, $p(x) = Cx^{-\alpha}$, where $C = e^{-1.4159} = 0.243$, $\alpha = 0.9550$ as measured in [61]. Thus, the average coin age in Bitcoin is $\text{avg}(p(x)) \approx 162$ days. For simplicity, for



(a) Communication costs (in PB) (b) Operational costs (in USD)

Figure 9: Stateless Blockchain case study. Communication and operational costs of a PSN for a given blockchain throughput and uptime for the bilinear accumulator.

a given stateless blockchain throughput y (in transactions per second (tps)), we treat that between each coin spending there are $y \cdot \text{avg}(p(x))$ new elements inserted into the accumulator representing the set of unspent coins in the cryptocurrency system. In case of the bilinear accumulator, we assume the system would use the longest available crs today, which has a length of 2^{28} [66]. Thus, the PSN can support the outsourcing of a membership proof update iff $y \cdot \text{avg}(p(x)) \leq 2^{28}$. In this case, the operational cost is $\text{cost} \cdot z \cdot (y \cdot \text{avg}(p(x))) \cdot t(\text{BlindMemProofUpd}(\cdot), y \cdot \text{avg}(p(x)))$ assuming that the PSN serves its clients for z days and the running times $t(\cdot, \cdot)$ were approximated from the measurements of Figure 8 using linear regression ($R^2 = 0.95$). Figure 9 suggests that operating a PSN with real-world workloads has low economic costs in the stateless blockchain application.

8 Open Problems and Future Directions

In this work, we presented efficient algorithms that allow an intermittently connected, resource-constrained client to privately delegate its (non-)membership proof updates to an untrusted server. We built and evaluated efficient schemes for the RSA and bilinear accumulators. We leave many open problems and foresee promising research directions for future work.

Other authenticated data structures Our primary focus in this work was to design private proof delegation for universal, dynamic accumulators. One could explore other authenticated data structures such as vector commitments (VC), polynomial commitment schemes, or authenticated dictionaries. For instance, we believe, similar techniques could work for the Hyperproofs VC scheme. The authors in [45] construct a key-value commitment (KVC) scheme in which a dictionary (a set of key-value pairs $D = \{(k_i, v_i)\}$) is represented directly as a sum of partial fractions. It seems in reach to design a private delegation scheme for the (non-)membership proofs in their KVC scheme with $O(1)$ server complexity.

Post-quantum privately delegatable accumulators We leave the construction of a post-quantum secure, privately delegatable universal, dynamic accumulator as a major open research question in this field [48].

Multi-server proof delegation It is promising to extend our protocols to a multi-server setting for better liveness. A basic design could be to have PSNs share a zero secret in the preprocessing phase, and collaboratively generate the proof update (blinded by a zero share). We leave the details of this protocol to future work.

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Setup(λ, n):

1. $(\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, p) \xleftarrow{\$} \text{GGen}(\lambda)$
2. $g_1 \xleftarrow{\$} \mathbb{G}_1, g_2 \xleftarrow{\$} \mathbb{G}_2, s_0(X) \leftarrow 1$
3. $\text{pp} := (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, g_1, g_2, s_0(X))$
4. $\text{crs} := (\{g_1^{\tau^i}\}_{i=0}^n, g_2, g_2^{\tau})$ for $\tau \xleftarrow{\$} \mathbb{F}_p$
5. **return** (pp, crs)

Add($\text{Acc}_t, S_t, s_t(X), s_j$):

1. **if** $s_j \in S_t$: **return** Acc_t
2. **else**:
3. $S_{t+1} \leftarrow S_t \cup \{s_j\}$
4. $s_{t+1}(X) \leftarrow s_t(X) \cdot (X - s_j)$
5. $\text{Acc}_{t+1} \leftarrow \text{KZG.Com}(s_{t+1}(X))$
6. **return** $\text{Acc}_{t+1}, S_{t+1}(X)$

Del(Acc_t, S_t, s_j):

1. **if** $s_j \notin S_t$: **return** Acc_t
2. **else**:
3. $S_{t+1} \leftarrow S_t \setminus \{s_j\}$
4. $s_{t+1}(X) \leftarrow s_t(X) / (X - s_j)$
5. $\text{Acc}_{t+1} \leftarrow \text{KZG.Com}(s_{t+1}(X))$
6. **return** $\text{Acc}_{t+1}, S_{t+1}(X)$

MemProofCreate($\text{Acc}_t, S_t, s_t(X), s_j$):

1. $q_{t,s_j}(X) \leftarrow s_t(X) / (X - s_j)$
2. $\pi_{t,s_j} \leftarrow \text{KZG.Com}(q_{t,s_j}(X))$
3. **return** π_{t,s_j}

NonMemProofCreate($\text{Acc}_t, S_t, s_t(X), s_j$):

1. $q_{t,s_j}(X) \leftarrow s_t(X) / (X - s_j)$
2. $y_{t,s_j} \leftarrow s_t(X) \bmod (X - s_j)$
3. $B_{t,s_j} \leftarrow \text{KZG.Com}(q_{t,s_j}(X))$
4. **return** $\overline{\pi_{t,s_j}} \leftarrow \{y_{t,s_j}, B_{t,s_j}\}$

MemVer($\text{Acc}_t, \pi_{t,s_j}, s_j$):

1. **return** 1 **if** $e(\pi_{t,s_j}, g_2^{\tau} / g_2^{s_j}) = e(\text{Acc}_t, g_2)$

NonMemVer($\text{Acc}_t, \overline{\pi_{t,s_j}}, s_j$):

1. $\{y_{t,s_j}, B_{t,s_j}\} \leftarrow \overline{\pi_{t,s_j}}$
2. $\text{lhs} \leftarrow e(B_{t,s_j}, g_2^{\tau-s_j}) e(g_1, g_2)^{y_{t,s_j}}$
3. **return** 1 **if** $\text{lhs} = e(\text{Acc}_t, g_2)$

Setup(λ):

1. $\mathbb{G} \xleftarrow{\$} \text{GGen}(\lambda)$
2. $g \xleftarrow{\$} \mathbb{G}$
3. **return** $\mathbb{G}, g$

Add(Acc_t, S_t, x):

1. **if** $x \in S_t$: **return** Acc_t
2. **else**:
3. $S_{t+1} \leftarrow S_t \cup \{x\}$
4. $u \leftarrow x$
5. $\text{Acc}_{t+1} \leftarrow \text{Acc}_t^x$
6. **return** Acc_{t+1}^x, u

Del(Acc_t, S_t, x):

1. **if** $x \notin S$: **return** Acc_t
2. **else**:
3. $S_{t+1} \leftarrow S_t \setminus \{x\}$
- 4a. $\text{Acc}_{t+1} \leftarrow g^{\prod_{s \in S_{t+1}} s}$
- 4b. $\text{Acc}_{t+1} \leftarrow \text{Acc}_t^{x^{-1}} \bmod |\mathbb{G}|$
5. $u \leftarrow \{x, \text{Acc}_t, \text{Acc}_{t+1}\}$
6. **return** Acc_{t+1}, u

MemProofCreate(Acc_t, S_t, x):

- 1a. $\pi_{t,x} \leftarrow g^{\prod_{s \in S_t, s \neq x} s}$
- 1b. $\pi_{t,x} \leftarrow \text{Acc}_t^{x^{-1}} \bmod |\mathbb{G}|$
2. **return** $\pi_{t,x}$

NonMemProofCreate(Acc_t, S_t, x):

1. $s^* \leftarrow \prod_{s \in S_t} s \bmod |\mathbb{G}|$
2. $a, b \bmod |\mathbb{G}| \leftarrow \text{Bezout}(s^*, x)$
3. $B \leftarrow g^b$
4. **return** $\overline{\pi_{t,x}} \leftarrow \{a, B\}$

MemVer($\text{Acc}_t, \pi_{t,x}, x$):

1. **return** 1 **if** $(\pi_{t,x})^x = \text{Acc}_t$

NonMemVer($\text{Acc}_t, \overline{\pi_{t,x}}, x$):

1. $\{a, B\} \leftarrow \overline{\pi_{t,x}}$
2. **return** 1 **if** $\text{Acc}_t^a B^x = g$

Figure 11: The algorithms of the bilinear accumulator [56].

Figure 10: The algorithms of the universal, dynamic RSA accumulator [9, 51]. Certain algorithms can be significantly sped up whenever the group order is known to a trusted accumulator manager (written in blue).

A Open Science

The implementation of our proposed private (non-)membership proof delegation protocols along with the accompanying benchmarks and test suites is open-sourced and can be accessed at <https://github.com/GlaszBoti/private-accumulator-proof-delegation>.

B Additional Background

B.1 The RSA and bilinear accumulators

We recall the algorithms of the universal, dynamic RSA [9, 51] and bilinear accumulators [30, 56] in Figures 10 and 11.

NIZK.Prove(pp, $\mathcal{R}_{\text{DLEq}}$, (g, u, h, v) , w):

1. $r \xleftarrow{\$} \mathbb{Z}_{2\lambda}$
2. $a \leftarrow g^r, \quad b \leftarrow h^r$
3. $e \leftarrow \text{H}_{\text{Primes}}(g, h, u, v, a, b)$
4. $z \leftarrow r + ew$
5. **return** $\pi := (a, b, z)$

NIZK.Verify(pp, $\mathcal{R}_{\text{DLEq}}$, (g, u, h, v) , π):

1. $(a, b, z) \leftarrow \pi$
2. $e \leftarrow \text{H}_{\text{Primes}}(g, h, u, v, a, b)$
3. **return** 1 if $g^z = a \cdot u^e \wedge h^z = b \cdot v^e$

Figure 12: A non-interactive zero-knowledge (NIZK) proof system for the relation $\mathcal{R}_{\text{DLEq}}$ using the Fiat-Shamir heuristic instantiated in a group of unknown order.

NIZK.Prove(pp, $\mathcal{R}_{\text{PoEEq}}$, (g, u, h, v) , crs, $\text{crs}_g, q^*(X)$):

1. $\tilde{g} \leftarrow \text{KZG.Com}(\text{crs}, q^*(X))$
2. $\alpha \leftarrow \text{H}(g, u, h, v)$
3. $h(X) \leftarrow q^*(X)/(X + \alpha)$
4. $\beta \leftarrow q^*(X) \bmod (X + \alpha)$
5. $Q_g \leftarrow \text{KZG.Com}(\text{crs}_g, h(X))$
6. $Q_h \leftarrow \text{KZG.Com}(\text{crs}, S_t(X) \cdot h(X))$
7. $\hat{g} \leftarrow \text{KZG.Com}(\text{crs}, h(X))$
8. **return** $\pi := (\tilde{g}, Q_g, Q_h, \hat{g}, \beta)$

NIZK.Verify(pp, $\mathcal{R}_{\text{PoEEq}}$, (g, u, h, v) , crs, crs_g, π):

1. $(\tilde{g}, Q_g, Q_h, \hat{g}, \beta) \leftarrow \pi$
2. $\alpha \leftarrow \text{H}(g, u, h, v)$
3. $eq_1 := e(Q_g, g_2^{\tau+\alpha}) \cdot e(g_1^\beta, g_2) \stackrel{?}{=} e(u, g_2)$
4. $eq_2 := e(Q_h, g_2^{\tau+\alpha}) \cdot e(g_1^\beta, g_2) \stackrel{?}{=} e(v, g_2)$
5. $eq_3 := e(\hat{g}, g_2^{\tau+\alpha}) \cdot e(g_1^\beta, g_2) \stackrel{?}{=} e(\tilde{g}, g_2)$
6. **return** 1 if $eq_1 \wedge eq_2 \wedge eq_3$

Figure 13: A non-interactive zero-knowledge (NIZK) proof system for the relation $\mathcal{R}_{\text{PoEEq}}$ using the Fiat-Shamir heuristic.

B.2 The KZG Polynomial Commitment scheme

KZG Polynomial Commitments. The Kate–Zaverucha–Goldberg (KZG) commitment scheme [47] allows committing to a polynomial over a finite field while later enabling succinct proofs of its evaluations.

$\text{KZG.Setup}(1^\lambda, n) \rightarrow \text{crs}$. Let $\tau \xleftarrow{\$} \mathbb{F}_p$. Compute the structured reference string $\text{crs} = (g_1, g_1^\tau, g_1^{\tau^2}, \dots, g_1^{\tau^n}, g_2, g_2^\tau)$. Output crs. These public parameters allow users to commit to polynomials of degree at most n without knowing the trapdoor τ itself.

$\text{KZG.Com}(\text{crs}, f) \rightarrow C$. Let $f(X) = \sum_{i=0}^d a_i X^i$ be a polynomial with $d \leq n$. Compute $C = g_1^{f(\tau)} = g_1^{\sum_{i=0}^d a_i \tau^i}$. Output

the commitment C . To commit to $f(X)$, the committer evaluates the polynomial at the hidden value τ in the exponent. Using the crs powers of τ , this exponentiation can be computed efficiently without knowing τ .

$\text{KZG.Open}(\text{crs}, f, z) \rightarrow \pi$. Compute the evaluation $y = f(z)$. Compute the quotient polynomial $q(X) = \frac{f(X) - y}{X - z}$. Compute the proof $\pi = g_1^{q(\tau)}$. Output (y, π) .

$\text{KZG.Vrfy}(\text{crs}, C, z, y, \pi) \rightarrow \{0, 1\}$. Accept if and only if $e(C/g_1^y, g_2) = e(\pi, g_2^z/g_2^\tau)$. Otherwise reject. The verifier checks a pairing equation encoding the identity $f(\tau) - y = (\tau - z)q(\tau)$ in the exponent.

Security properties. The KZG commitment scheme satisfies correctness, (polynomial) binding, and evaluation binding. Intuitively, polynomial binding ensures that a polynomial commitment cannot be opened to two different polynomials. Evaluation binding ensures that at any point $z \in \mathbb{F}_p$, the prover can only open the committed polynomial to a single evaluation point y . Finally, correctness ensures that an honest opening proof is always accepted by the verifier. For the formal security definitions, the reader is referred to [47].

B.3 NIZKs for NP-statements in Groups of Unknown Order

Discrete logarithm equality (sometimes Chaum–Pedersen proof) (DLEq) [14, 23].

$$\mathcal{R}_{\text{DLEq}} = \{((g, u, h, v) \in \mathbb{G}^4); w \in \mathbb{Z} : u = g^w \wedge v = h^w\}. \quad (\text{B.1})$$

C Deferred Security Proofs

C.1 RSA membership proof delegation

Theorem 6 (RSA accumulator is privately delegatable for membership proofs). Our construction in Figure 4 yields a secure, privately delegatable membership proof update scheme for the RSA accumulator under the strong RSA assumption and knowledge soundness of Figure 12.

PROOF. Correctness. For any accumulated set S_t and any element $x \in S_t$, an honestly run proof update should yield a valid membership proof for the updated accumulator $\text{Acc}_{t'}$ that is accepted with overwhelming probability.

Let the accumulator at time t be $\text{Acc}_t = g^{\prod_{s \in S_t} s}$. For $x \in S_t$ the corresponding membership proof is $\pi_{t,x} = g^{\prod_{s \in S_t / \{x\}} s}$. Then, an honestly generated proof update on the blinded witness $\pi'_{t,x} = \pi_{t,x} \cdot g^r$ yields

$$(\pi'_{t,x}, \text{aux}) = ((\pi'_{t,x})^{\prod_{i=1}^k x_i}, ((\pi'_{t,x})^r, g^r, z)), \quad (\text{C.1})$$

where the current set of accumulated values is $S_{t'} = (S_t \cup \{x_i\}_{i=1}^k)$. The unblind algorithm outputs

$$\pi_{t',x} = \frac{\pi'_{t,x}}{(g^\Delta)^r} = \pi_{t,x}^\Delta. \quad (\text{C.2})$$

Finally, we verify the unblinded, updated membership proof

$$(\pi_{t',x})^x = \left(g^{\prod_{s \in S_t / \{x\}} s}\right)^{x\Delta} = g^{(\prod_{s \in S_t} s) \cdot \Delta} = \text{Acc}_{t'}, \quad (\text{C.3})$$

which always leads to acceptance, thus completing the proof of correctness.

Blinded Membership Proof Indistinguishability. Consider the blinded membership proof indistinguishability game in Figure 3. The adversary $\mathcal{A}$ sends x_0, x_1 , after which the challenger C samples a random bit $b \xleftarrow{\$} \{0, 1\}$ and responds with the blinded membership proof $\pi'_{1,x_b} = \pi_{1,x_b} \cdot g^r$, where r is freshly sampled for each query. If r is sampled uniformly at random, then $\pi_{1,x_b} \cdot g^r$ is uniform in $\mathbb{G}$, and thus, independent of the challenger's bit b . In other words, we have information-theoretic indistinguishability. Therefore,

$$\text{Adv}_{\mathcal{A}}^{\text{ind}} = \left| \Pr[b' = b] - \frac{1}{2} \right| \leq \text{negl}(\lambda).$$

Delegation Soundness. Towards contradiction, let us assume that there exists an adversary $\mathcal{A}$ that is able to output a blinded membership proof $(\pi'_{t',x}, \text{aux})$ proof for an element $x \in S_t$ with non-negligible probability s.t.,

$$\begin{aligned} & \text{MemVer}(\text{UnBlindProof}(\text{Acc}_t, \text{st}, \pi'_{t',x}) = \\ & 0 \wedge \text{VerBlindProofUpd}(\text{pp}, \text{Acc}_{t'}, \pi'_{t,x}, \pi'_{t',x}, \text{aux}) = 1. \end{aligned}$$

Since the applied NIZK for the discrete logarithm equality NP-relation is an argument of knowledge, the extractor $\mathcal{E}$ outputs the triplet (a, b, z) satisfying the NIZK.Verify function with non-negligible probability. Thus, there exists a value Δ s.t. $(g, \text{Acc}_t, \pi'_{t,x}) = (g^\Delta, \text{Acc}_t^\Delta, (\pi'_{t,x})^\Delta)$. Unblinding the blinded RSA membership proof update yields

$$\pi'_{t',x} = \frac{\pi'_{t',x}}{B^r}.$$

Substituting this into the MemVer algorithm, we have $\pi'_{t',x} = \text{Acc}_{t'}$ as desired. Thus, the only way an adversary could produce a valid BlindProofUpd that leads to an invalid unblinded proof is by breaking the knowledge soundness of the applied NIZK. $\square$

C.2 RSA non-membership proof delegation

Theorem 7 (The RSA accumulator is privately delegatable for non-membership proofs). Our construction in Figure 5 yields a secure, privately delegatable non-membership proof update scheme for the RSA accumulator under strong RSA and factoring assumption in the ROM.

PROOF. Correctness. For $x \notin S_t$, we have $\overline{\pi'_{t',x}} = \{a, g^b\}$, for $(a, b) \leftarrow \text{Bezout}(s^*, x \cdot q)$. The unblinded non-membership proof is obtained by $B' := (g^b)^q$, while the Bézout-coefficient a remains unchanged. By construction, $as^* + b(xq) = 1$. Due to the commutativity of integer multiplication, we also have $as^* + bq(x) = 1$. Thus, (a, bq) are the Bézout-coefficients of s^* and x , as required.

Weak Blinded Non-Membership Proof Indistinguishability. Let us assume that there is an efficient adversary $\mathcal{A}$ that on input $N := \overline{\pi'_{t,x}}$ outputs with non-negligible probability $\frac{1}{\text{poly}(\lambda)}$ to which element $x \in \{0, 1\}^\lambda$ the blinded non-membership proof belongs, i.e., $N = x \cdot q$ for some primes $x, q \in \text{Primes}(\lambda)$.

Adversary $\mathcal{A}$ can be used by another machine $\mathcal{B}$ to break the factoring assumption. Machine $\mathcal{B}$ on any input N should just forward its input to $\mathcal{A}$ who factors it, by assumption, with non-negligible probability. Adversary $\mathcal{B}$ clearly has the same time-complexity and success probability as $\mathcal{A}$ in its own weak blinded non-membership proof indistinguishability game.

Delegation soundness. Towards contradiction, assume that there exists an adversary $\mathcal{A}$ that outputs $x \notin S_t$ and a blinded non-membership proof $\overline{\pi'_{t',x}} = \{a, B\}$ for the updated accumulator $\text{Acc}_{t'}$ such that

$$g = \text{Acc}_{t'}^a \cdot B^y$$

holds in the blinded verification (for some exponent y used in the blinded update protocol), but the RSA accumulator's non-membership proof verification fails for the corresponding unblinded proof, i.e.,

$$g \neq \text{Acc}_{t'}^a \cdot (B^q)^x.$$

However, by construction of the blinded update, we have $y = q \cdot x$. Substituting this into the first equation yields

$$g = \text{Acc}_{t'}^a \cdot B^{qx} = \text{Acc}_{t'}^a \cdot (B^q)^x,$$

which directly contradicts the inequality above. Hence, both equations cannot hold simultaneously with non-negligible probability, and no such adversary $\mathcal{A}$ can exist. $\square$

C.3 Bilinear membership proof delegation

Theorem 8 (Bilinear accumulator is privately delegatable). The construction in Figure 6 is a secure, privately delegatable membership proof update scheme for the bilinear accumulator under q -SDH in the algebraic group model and ROM.

PROOF. Correctness. By the definition of the blinding algorithm, we have that $\text{crs}'_j = \{\text{KZG.Com}(rX^i q(X))\}_{i=1}^k$. Let us denote the polynomial of the newly added elements $\{x_i\}_{i=1}^k$ in the monomial bases as $q^*(X) = \prod_{i=1}^k (X - x_i) = \prod_{j=0}^k a_j X^j$. Using the homomorphic properties of KZG, in the BlindProofUpd($\cdot$) algorithm, we have

$$\begin{aligned} \pi'_{t',s_j} &= \prod_{j=0}^k (\text{crs}'_j)^{a_j} = \prod_{j=0}^k \text{KZG.Com}(rX^j q(X))^{a_j} = \\ &= \prod_{j=0}^k \text{KZG.Com}(a_j \cdot rX^j q(X)) = \text{KZG.Com}\left(\sum_{j=0}^k a_j \cdot X^j r q(X)\right), \end{aligned}$$

factoring out the update independent polynomial $r q(X)$ yields the following.

$$\begin{aligned} \text{KZG.Com}\left(\sum_{j=0}^k a_j \cdot X^j r q(X)\right) &= \text{KZG.Com}\left(r q(X) \cdot \sum_{j=0}^k a_j X^j\right) = \\ &= \text{KZG.Com}(r q(X) q^*(X)). \end{aligned}$$

By the unblinding algorithm, $\text{UnBlindProof}(\cdot)$, we have that

$$\begin{aligned} (\pi'_{t',s_j})^{-r} &= \text{KZG.Com}(r q(X) q^*(X))^{-r} = \\ &= \text{KZG.Com}(q(X) q^*(X)) = \pi_{t',s_j}, \end{aligned}$$

as desired. Therefore, π_{t',s_j} is indeed the KZG commitment to the new quotient polynomial $q(X) q^*(X)$.

Blinded Membership Proof Indistinguishability. In the blinded membership proof indistinguishability game, the adversary chooses s_0, s_1 and sees a crs randomized by the challenger with $r \in_R \mathbb{F}_p$, i.e., $\text{crs}' := \{\text{KZG.Com}(rX^i q_b(X))_{i=1}^k\}$, where $q_b(X) = \frac{f(X)}{X-s_b}$, for $b \in \{0, 1\}$. Since $q_b(\tau) \neq 0$, let $s := r \cdot q_b(\tau)$. Since in both cases s is uniform, i.e., $s \sim U(\mathbb{F}_p)$, and multiplication by a non-zero constant in $\mathbb{F}_p$ is a permutation, in both cases, the adversary sees the distribution $\{g^{s\tau^i}\}_{i=1}^k$ for a secret $s \in \mathbb{F}_p$. Thus, regardless of which element has been chosen by the challenger, $\mathcal{A}$ sees perfectly indistinguishable distributions $\{g^{s\tau^i}\}_{i=1}^k \approx \{g^{s^*\tau^i}\}_{i=1}^k$ for the challenger chosen $(s, s^*) \in_R \mathbb{F}_p^2$.

Delegation Soundness. The NIZK proof system applied for the NP-language $\mathcal{R}_{\text{PoEEq}}$ was shown to be knowledge sound in the algebraic group model [63]. Thus, if aux is verified for the statement of the polynomial exponentiation equality $\mathcal{R}_{\text{PoEEq}}$, then for the public instance $(\pi'_{t,s_j}, \pi'_{t',s_j}, \text{Acc}_t, \text{Acc}_{t'})$, there is a polynomial $q^*(X) \in \mathbb{F}_p^{\leq k}[X]$ s.t.

$$(\pi'_{t,s_j})^{q^*(\tau)} = \pi'_{t',s_j} \wedge (\text{Acc}_t)^{q^*(\tau)} = \text{Acc}_{t'}, \quad (\text{C.4})$$

where τ is the trapdoor of the KZG crs. We see in Equation (C.4) that $q^*(X)$ encodes the update polynomial $\prod_{i=1}^k (X - x_i)$ between the two accumulator digests Acc_t and $\text{Acc}_{t'}$. Thus,

$$(\pi'_{t',s_j})^{r^{-1} \bmod p} = (g^{r q(\tau) q^*(\tau)})^{r^{-1} \bmod p} = \pi'_{t',s_j} \quad (\text{C.5})$$

is a valid membership proof as required. $\square$

C.4 Bilinear non-membership proof delegation

Theorem 9 (Bilinear accumulator is privately delegatable for non-membership proofs). Our construction in Figure 7 yields a secure, privately delegatable non-membership proof update scheme for the bilinear accumulator under q -SDH in the ROM.

PROOF. Correctness. First, the client generates a short randomized common reference string $\text{crs}' := \{g^{r_1 q_{t,s_j}(\tau)}, g^{r_1 \tau q_{t,s_j}(\tau)}\}$ which is used by the server to KZG-commit to the polynomial $X - s_{n+1}$, i.e., $Q' := \text{KZG.Com}(\text{crs}', X - s_{n+1})$. Then, in the $\text{UnBlindNonMemProof}(\cdot)$ algorithm, the client rerandomizes Q' and obtains $Q = (Q')^{r_1^{-1} \bmod p} \cdot g^{y_{t,s_j}} = g^{\tau q_{t,s_j}(\tau) - s_{n+1} q_{t,s_j}(\tau) + y_{t,s_j}} = \text{KZG.Com}(\text{crs}, (X - s_{n+1})q_{t,s_j}(X) + y_{t,s_j})$, which is exactly the desired quotient polynomial for the updated non-membership proof, cf. Equation (5.3). Similarly, the client blinds y_{t,s_j} to $y'_{t,s_j} = r_2 \cdot y_{t,s_j}$. In turn, the server computes $y'_{t+1,s_j} = s_{n+1} \cdot y'_{t,s_j}$ which is derandomized at the end of the protocol by the client in the $\text{UnBlindNonMemProof}(\cdot)$ algorithm as $y_{t,s_j} s_j - y'_{t+1,s_j} / r_2 = y_{t,s_j} s_j - s_{n+1} \cdot y'_{t,s_j} / r_2 = y_{t,s_j} s_j - s_{n+1} \cdot r_2 \cdot y_{t,s_j} / r_2 = y_{t,s_j} s_j - s_{n+1} y_{t,s_j}$, which is the desired scalar element required by the updated non-membership proof recurrence relation, cf. Equation (5.3).

Blinded Non-Membership Proof Indistinguishability. Given two blinded membership proofs $\overline{\pi'_{t,s_0}} = \{\text{crs}'_0, r_{0,2} \cdot y_{t,s_0}\}$ and $\overline{\pi'_{t,s_1}} = \{\text{crs}'_1, r_{1,2} \cdot y_{t,s_1}\}$ for $\text{crs}'_0 := \{g^{r_{0,1} q_{t,s_0}(\tau)}, g^{r_{0,1} \tau q_{t,s_0}(\tau)}\}$ and $\text{crs}'_1 := \{g^{r_{1,1} q_{t,s_1}(\tau)}, g^{r_{1,1} \tau q_{t,s_1}(\tau)}\}$, it is clear that $\overline{\pi'_{t,s_0}}$ is information-theoretically indistinguishable from $\overline{\pi'_{t,s_1}}$, since, first, $r_{0,2} \cdot y_{t,s_0} \approx_s r_{1,2} \cdot y_{t,s_1}$ for uniformly random $(r_{0,2}, r_{1,2}) \leftarrow \mathbb{F}_p^2$, and second, $\text{crs}'_0 \approx_s \text{crs}'_1$, because $r_{0,1}$ and $r_{1,1}$ act as random permutations on crs_0 and crs_1 .

Delegation Soundness. Consider any PPT adversary (PSN) that, given $\overline{\pi'_{t,s_j}}$ and the public update $(s_{n+1}, \text{Acc}_t, \text{Acc}_{t+1})$, outputs $(\overline{\pi'_{t+1,s_j}}, \text{aux})$ such that $\text{VerBlindNonMemProofUpd}(\overline{\pi'_{t,s_j}}, \overline{\pi'_{t+1,s_j}}, s_{n+1}, \text{Acc}_{t+1}, \text{aux}) = 1$. We show that, except with negligible probability, the unblinded witness $\overline{\pi'_{t+1,s_j}}$ returned by $\text{UnBlindNonMemProof}$ is a valid non-membership proof for $s_j \notin S_{t+1}$ under Acc_{t+1} . The pairing check can be rewritten, using bilinearity and $\text{Acc}_{t+1} = \text{Acc}_t^{\tau - s_{n+1}}$, as an equality of exponents in $\mathbb{G}_1$ linking Q'_t and Q'_{t+1} via the same factor $(\tau - s_{n+1})$. By soundness of the accumulator and binding of KZG commitments, any deviation of Q'_{t+1} from the honest updated value (i.e., the commitment to $r_{1,q_{t+1,s_j}}(\tau)$, where $q_{t+1,s_j}(X) = (X - s_{n+1})q_{t,s_j}(X) + y_{t,s_j}$) would violate this pairing equation except with negligible probability. Finally, in $\text{UnBlindNonMemProof}$ the client computes

$$Q := (Q'_{t+1})^{r_1^{-1}} \cdot g^{y_{t,s_j}} = g^{q_{t+1,s_j}(\tau)},$$

so $\overline{\pi'_{t+1,s_j}} = (Q, y_{t+1,s_j})$ is exactly the standard non-membership witness for $s_j \notin S_{t+1}$ under Acc_{t+1} . Therefore, the probability that a PPT adversary produces an accepting transcript for $\text{VerBlindNonMemProofUpd}$ while the resulting unblinded witness is invalid is negligible. $\square$

C.5 Server-side Batching

The following theorem states that asymptotic server-side batching for our *membership proof delegation* protocols (cf. Figures 4 and 6), i.e., $O(n)$: sublinear in the number of delegation requests, is impossible in the generic group model. Specifically, we assume there are n users trying to update their membership proofs for the same transition $t \rightarrow t'$. Then, the membership proof update, at a high level, translates to exponentiating n random group $\{g_i\}_{i=1}^n$ with the same exponent x . As we show next, this computation needs $\Omega(n)$ exponentiation.

Lemma 10 (Linear lower bound for component-wise exponentiation in the generic group model). Let $\mathbb{G}$ be a cyclic group of prime order p in the generic group model. Let $g_1, \dots, g_n \in \mathbb{G}$ be independent random group elements and let $x \in \mathbb{Z}_p$ be known. Any generic algorithm that outputs

$$(g_1^x, g_2^x, \dots, g_n^x)$$

must perform at least n oracle queries. In particular, the generic complexity of this task is $\Omega(n)$.

PROOF. Fix a generator g of $\mathbb{G}$ and write

$$g_i = g^{\alpha_i},$$

where $(\alpha_1, \dots, \alpha_n) \xleftarrow{\$} \mathbb{Z}_p^n$ are independent, uniform random exponents unknown to the algorithm. By the standard analysis of generic algorithms [62], every group element computed by a generic algorithm can be written as

$$g^{L(\alpha_1, \dots, \alpha_n)}$$

where L is a linear polynomial

$$L(\alpha_1, \dots, \alpha_n) = \sum_{j=1}^n a_j \alpha_j$$

with coefficients $a_j \in \mathbb{Z}_p$ known to the algorithm.

```

BlindBatchRSAMemProof(pp,  $\pi_t, \{s_j\}_{j=1}^m$ ):
1.  $st := r \xleftarrow{\$} \mathbb{Z}_{2\lambda}, B := \{s_j\}_{j=1}^m \subseteq S$ 
2.  $\pi'_{t,B} := \pi_{t,B} \cdot g^r$ 
3. return ( $\pi'_{t,B}, st$ )

BlindBatchRSANonMemProof(pp,  $\{s_j\}_{j=1}^m$ ):
1. if  $\{s_j\}_{j=1}^m \cap S_t \neq \emptyset$  : return  $\perp$ 
2.  $B := \prod_{j=1}^m s_j$ 
3.  $st := q \xleftarrow{\$} \text{Primes}(1^\lambda)$ 
4.  $\overline{\pi'_{t,B}} := B \cdot q$ 
5. return ( $\overline{\pi'_{t,B}}, st$ )

BlindBatchBilinearMemProof(pp,  $\text{Acc}_t, S_t, \{s_j\}_{j=1}^m, \pi_t, \{s_j\}_{j=1}^m$ ):
1.  $st := r \xleftarrow{\$} \mathbb{F}_p, B := \{s_j\}_{j=1}^m \subseteq S$ 
2.  $q(X) \leftarrow \prod_{i=1, s_i \notin B}^n (X - s_i)$ 
3.  $\text{crs}'_B \leftarrow \{\text{KZG.Com}(\text{crs}, rX^i q(X))\}_{i=0}^k$ 
4.  $\pi'_{t,B} \leftarrow \{\text{crs}'_B\}$ 
5. return ( $\pi'_{t,B}, st$ )

```

Figure 14: The batch delegation algorithms for privately updating batch (non-)membership proofs in the RSA and bilinear accumulators. The rest of the algorithms remain the same as in the non-batch setting, i.e., cf. Figures 4 to 6.

Thus, each known group element corresponds to a vector

$$a = (a_1, \dots, a_n) \in \mathbb{Z}_p^n.$$

Let $V_t \subseteq \mathbb{Z}_p^n$ denote the linear span of all coefficient vectors corresponding to elements known by the algorithm after t oracle queries.

Initially the algorithm knows $g_1, \dots, g_n$, corresponding to the vectors

$$e_1, \dots, e_n,$$

where e_i is the i -th standard basis vector. Hence

$$V_0 = \text{span}\{e_1, \dots, e_n\}.$$

Each oracle query (group multiplication or inversion) produces at most one new group element, whose coefficient vector is a linear combination of previously known vectors. Consequently, after t oracle queries the algorithm knows at most $n + t$ group elements.

The desired outputs are

$$g_i^x = g^{x\alpha_i},$$

which correspond to the vectors

$$xe_1, \dots, xe_n.$$

Since the α_i are independent random variables, these elements are independent group elements. Therefore, the algorithm must explicitly produce the encoding of each g_i^x .

As each oracle query can produce at most one new group element, producing n distinct outputs requires at least n oracle queries. Thus, any generic algorithm computing $(g_1^x, \dots, g_n^x)$ requires at least n oracle queries, proving the $\Omega(n)$ lower bound. $\square$

It would be fascinating to prove lower bounds for the server complexity of updating n client *non-membership proofs* in the standard or even just in the generic or algebraic group models. We leave this seemingly harder problem to future work.

D Batching Proof Delegation

The accumulators we study in this work support constant-sized batch (non-)membership proofs [14, 63]. These succinct batch proofs attest to the (non-)inclusion of a batch B of elements, such that $B \subseteq S_t$. Thanks to the algebraic properties of these batch proofs, it is straightforward to modify our proof update delegation protocols to the batch setting, cf. Figure 14. Essentially, the same blinding technique works that we developed in Sections 5.3 and 5.4.1. More precisely, we only need to change the semantics of the Blind(Non)MemProof($\cdot$) algorithms, i.e., these algorithms work correctly even if we input batch (non-)membership proofs instead of single (non-)membership proofs. Next, we formalize this intuition.

E Extended Performance Evaluation

Next, we detail some performance measurements that we could not include in the main text due to space constraints.

E.1 Constant-Time Algorithms

In Table 4, we provide benchmarks for the constant-time algorithms, Blind(Non)MemProof($\cdot$), UnBlind(Non)MemProof($\cdot$) and Blind(Non)MemProofUpd($\cdot$), i.e., exactly the algorithms run by the intermittently connected, resource-constrained clients. Observe in Table 4 that all client algorithms run in the millisecond range, showcasing the practicality of our protocols for clients, one of our initial design goals. Moreover, the results show that the bilinear accumulator-based constructions are significantly faster, with sub-millisecond running time for most operations. In contrast, the RSA accumulator instantiated with class groups inherently avoids trusted setups but incurs substantial overhead for all our algorithms due to larger group sizes and the slower Abelian group law for composing binary quadratic forms. The low variance across all schemes and algorithms provides evidence of stable and deterministic behavior of our implementations.

E.2 Runtime Breakdown of (Non-)Membership Proof Updates

Figure 15 illustrates what percentage of the runtime is spent on various cryptographic operations, i.e., hashing, exponentiation (multi-scalar multiplication (MSM) in case of the bilinear accumulator), and NIZK proving during private (non-)membership proof updates when updating for a single element (i.e., $k = 1$). We note that in the case of private membership proof updates, the size of the exponent grows linearly in

Algorithm	RSA		Class Group		Bilinear	
	Avg(ms)	Var(ms)	Avg(ms)	Var(ms)	Avg(ms)	Var(ms)
BlindMemProof($\cdot$)	18.82	4.60	56.68	15.37	0.94	0.00
UnBlindMemProof($\cdot$)	10.71	0.00	46.58	0.00	0.21	0.00
BlindNonMemProof($\cdot$)	13.15	0.60	8.03	0.69	0.75	0.00
UnBlindNonMemProof($\cdot$)	6.25	0.00	40.82	0.01	0.32	0.00
VerBlindMemProofUpd($\cdot$)	90.95	0.01	513.39	2.65	8.23	0.00
VerBlindNonMemProofUpd($\cdot$)	20.29	0.01	161.65	0.30	8.31	0.00

Table 4: Performance evaluation of the constant-time algorithms in the private (non-)membership proof update protocols for the RSA and bilinear accumulators

k (cf. Section 7.1), making exponentiation the dominant operation. For all instantiations of the RSA accumulator, we expect that, as $k \rightarrow \infty$, the dominant cost will be the exponentiation costs. Luckily, specialized hardware chips that can compute exponentiation in groups of unknown order (*i.e.*, in RSA and class groups) extremely fast are already available due to verifiable delay function deployments [12]. These hardware chips could be leveraged by PSNs for these computational tasks.

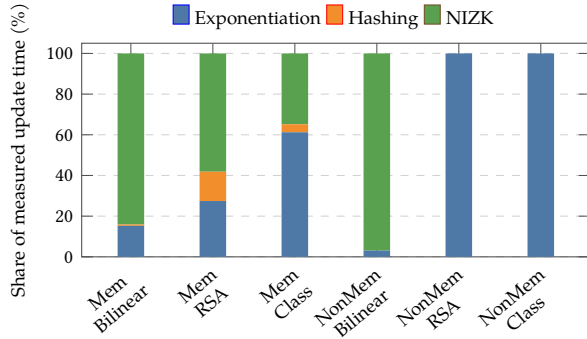
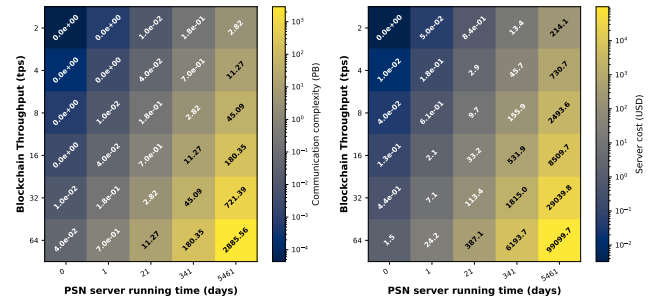


Figure 15: Runtime breakdown in terms of exponentiation, hashing and NIZK proving of (non-)membership proof updates when updating for a single element.

E.3 Case Study: PSN for Stateless Blockchains from the RSA accumulator

We study the operation and communication costs of deploying a PSN for the stateless blockchain use case, whenever the unspent coins are accumulated using the *trapdoorless* RSA accumulator, cf. Figure 16. Observe in Figure 16 that the RSA

membership proof delegation protocol that we developed can support arbitrarily scalable stateless blockchains, since there is no public parameter that would limit the desired proof update functionality. However, we also see that the RSA proof delegation algorithms incur higher communication and operational costs to the PSN server compared to the bilinear accumulator case (again, proof delegation running times for larger update sizes were approximated from Figure 8a using linear regression). We believe that the measured operational costs are reasonable for a real-world PSN deployment. For example, running a PSN server would cost ≈ 100 K USD when updating RSA membership proofs for the users of a stateless blockchain with 64 transaction per second for an entire year. On the other hand, these numbers translate into $\approx 5.94 \times 10^{-10}$ USD cost for the client for each transaction submitted, which we can consider to be an economically negligible cost.



(a) Communication costs (in PB) (b) Operational costs (in USD)

Figure 16: Stateless Blockchain case study. Communication costs (in PB) and operational costs (in USD) of a PSN server for a given stateless blockchain throughput and uptime for the RSA accumulator.